

Generating synthetic interview transcripts in a psychiatric setting using a double-language model approach.

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Abstract

Training data is a key part of producing and improving large language models (LLMs). Narrative interviews form the basis of mental health treatment plans, and history-taking is an ideal application for LLMs. However, the use of history-taking training data is restricted by privacy and legal considerations making synthetic forms an ideal alternative. Our tool uses two LLMs to create a synthetic interview transcript in a clinical intake setting (the double-model approach) and assume both interviewer and patient roles. A double model can counteract the effects of generating an interview from one model, where biases and context size limitations narrow the diversity of patient characteristics and assistant responses. We implement the double-model tool in Python, creating two instances of ChatGPT 4 with the OpenAI API in a Jupyter Notebook format. The user can customize a question bank and create patients with randomized attributes. Additionally, we embed edge case situations—for example, if the patient neglects to provide their medication dosage, the assistant should ask an appropriate follow up question. The quality of the synthetic interviews was evaluated using metrics of interview completeness, accuracy, and safety using established LLM benchmarks and in general observed improved metrics with the double model. The performance of a model trained on our interview dataset against an untrained model was used to evaluate how consistently responded correctly to edge cases. Our double-model approach can be applied for generating additional conversational datasets and investigating LLM behavior.

GitHub Code repos: https://github.com/ubcbraincircuits/transcript_generation

Single and double model + code repos

Introduction

Keeping an accurate record of patient history is essential to holistic diagnosis and care, especially in psychiatry. The foundation of a patient's background is established during the initial consultation. Yet, clinical care surveys show that documentation takes up nearly half of clinical time (Sinsky et al., 2016), correlating with physician burnout and dissatisfaction (Shanafelt et al., 2016). To reduce this clerical burden, we propose the application of an AI “chatbot” assistant to help clinicians gather and maintain background information.

The emergence of large language models (LLMs) in machine learning offers a potential solution. LLMs possess the ability to process and generate natural language without the need to manually define responses or behaviors. Trained on massive amounts of text datasets, LLMs gain a generalized understanding of language across different contexts. Further reinforcement of certain speech patterns or domain-specific medical knowledge is possible through additional task-specific training, a process known as “fine-tuning”.¹² These models could overcome limitations faced by current conversational agents in healthcare.

Given the promising possibilities that computer-automated history-taking and artificial intelligence have for improving patient care and reducing clinician strain, there are still challenges to implementation. Privacy is a major concern for clinicians regarding the use of AI tools due to a lack of transparency in how patient data is used by proprietary companies like OpenAI's ChatGPT.¹³ Due to the sensitive nature of the information being collected, numerous precautions need to be considered while attempting to privately systematize the digital AI tool. There is a need for collaboration between clinical practitioners and AI developers to fully realize the potential that this emerging technology can fulfill. Our project's intention is not for artificial intelligence to supplant the role of a psychiatrist, but to augment the psychiatric practice by establishing workflows in particular those that utilize synthetic interviews and a double model approach and some standards by which next generation chatbots can be created.

The double model interview generator is a pipeline in which one can create an interview transcript between two chatbots. This method consists of a modifiable set of text files to adjust the prompt which each chatbot receives (`assistant_prompt.txt` and `user_prompt.txt`) and the aspects of the interview that the interviewer bot must cover (`questionbank.txt`). This allows for interviews to be generated with greater variety, and in different contexts. For example, the prompts can be changed to create a transcript for a job interview by altering the user prompt to contain certain qualifications and the interviewer prompt to ask directed questions. Another example could be in legal interviews, where key witnesses may need to provide a testimony. We

want to create transcripts that can be used as training data for a smaller local language model for privacy reasons.

Results

Transcripts

We applied the SM and DM methods using 10 synthetic patients, resulting in n=20 transcripts. The patient is also assigned either a “rambling” or “brief” typing style and instructed to misspell a medication name as an edge case scenario.

Functionality

DM patients with the “rambling” typing style had significantly higher average message lengths ($P < .001$, Fig 3a) and significantly lower Distinct-1 scores ($P < .001$, Fig 3b) compared with SM “rambling” patients.

There were significant differences between assistants' coverage scores with the DM and SM method ($P < .001$, Fig 4a).

Edge Case Scenario

All patients instructed to misspell the medication name did so (“hit”). While most assistants responded appropriately, a minority did not correct the patient as instructed (“false negative”) (Fig 4b).

Assistant Details

Assistant sentences were mostly empathy statements in all the transcripts, with in-depth questions as second most common (Fig 4c).

Conclusion

- Results suggest that there are noticeable differences in the transcript depending on generation model used
- DM can be used when longer transcripts are necessary, SM is suitable for shorter interactions
- Prompts, question bank, and patient randomization can be easily edited to suit different goals/scenarios
- Our transcript generation pipeline is a straightforward method to create and examine a variety of synthetic interview transcripts for finetuning purposes

Figure 1. Operational Steps for Synthetic Interview Generation
Synthetic Interview
Transcript generation is grouped into three main steps with associated Jupyter Notebook files, available on our GitHub at: github.com/ubcbraincircuits/transcript_generation

Figure 2. Single Model (SM) and Double Model (DM) Diagrams

In the SM approach, one GPT-4 model creates a transcript based on both the Question Bank and a Patient Profile.

In the DM approach, two GPT-4 models create the transcript by interacting with each other. Only the “Assistant” model has access to the Question Bank, and only the “Patient” model has access to the Patient Profile.

Both the SM and DM use the same “Patient” and “Assistant” prompts. Our prompts were built upon templates used in previous studies.⁴

Figures 3a, 3b

Transcript

Analysis

Figures were generated by separating the transcript into each role's messages.

3a measures each role's average message length in characters.

3b shows the average "Distinct-1" score, which is

the number of
unique words
over all words in
the role's
messages and is
a measure of
colloquial
speech.⁴

Figures 4a-c GPT-Evaluated Transcript Analysis

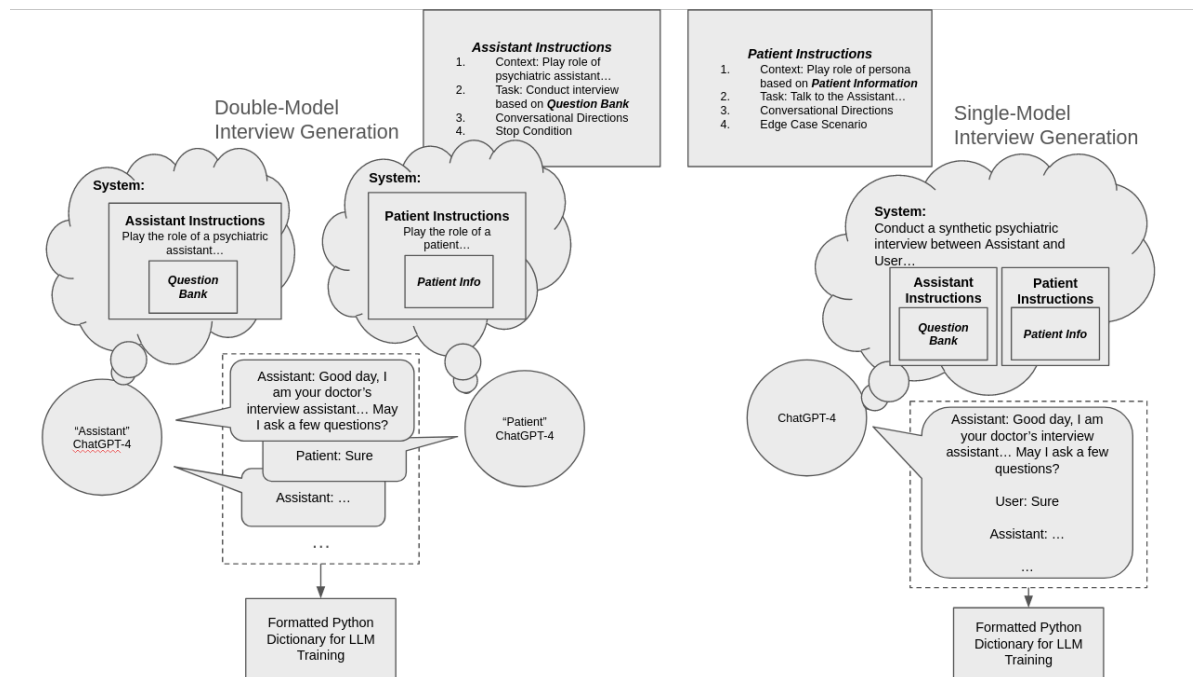
4a shows the GPT-evaluated
"coverage score", indicating degree of
overlap between the transcript and
the Patient Profile or Question Bank.
4b shows whether the edge case is
executed in the transcript or not. The
three categories are "hit", "false
positive", or "false negative".
4c shows the average breakdown of
the Assistant's messages into
sentence categories.

Discussion

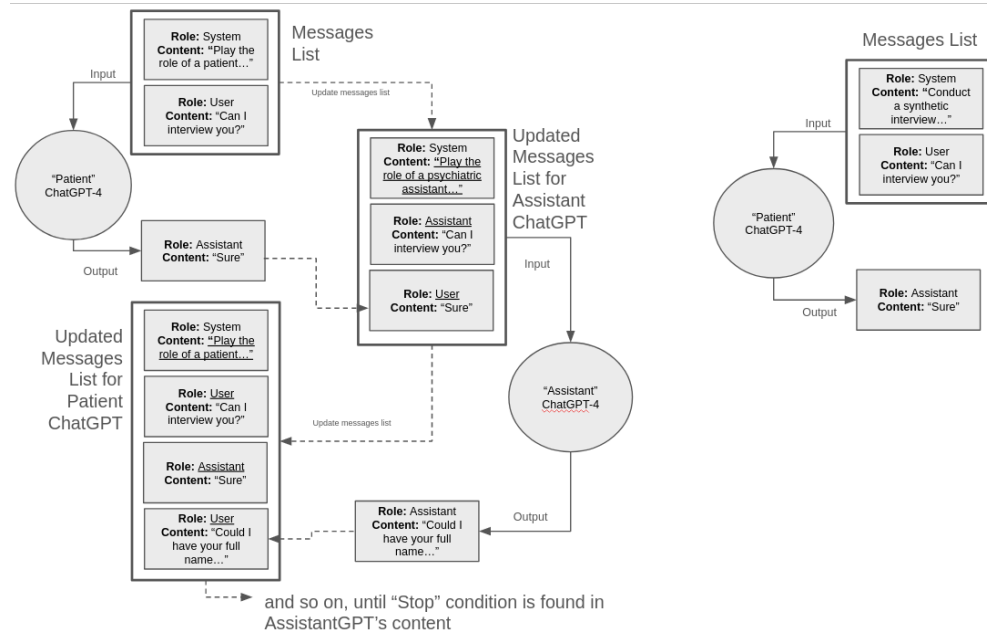
We have created a tool to generate synthetic patients and medical histories as a means for users to generate training data for locally trained models that may help preserve patient privacy. This in particular pertains to patient privacy laws to protect sensitive personal information in the case of psychiatry. We deploy the double model dataset generation process for psychiatric patient intake interviews. As the quantity and quality of training data dictates chatbot performance, we wish to create realistic transcripts that reflect interactions with a variety of patient backgrounds. The most important considerations are that the transcripts can incorporate edge case handling naturally in the script itself, and that appropriate in-depth information arises from this procedure. appropriately detailed and specific information arises from a combination of the assistant's ability to ask not only the questions in the list, but also appropriate follow-up questions that pinpoint missing details. it also depends on the user's ability to come up with further details consistent with the basic profile it has been provided.

When we generate a summary of the transcript, we should be able to draw out the similarities between the input patient prompt and the history that arises from the interview process. (look up

philosophy of interviews?) we should also be able to compare the summaries created from the single model and double model generation methods to rank the detailedness and creativity of each transcript.



code structure for contacting LLMs



Code Architecture

Background/Related Works

LLMs in Clinical Medicine/Psychiatry

- use of LLMs in zero-shot evaluations
 - ClinicalBERT: hospital readmission predictions improve after fine-tuning on clinical notes
- use of LLMs for interviews
 - LLMs can be used to tag and summarize patient behavior from interviews; eventual goal is for model to provide a second opinion in doctor-patient conversations/for diagnoses (So et al., 2024)
 - LLMs can be used to generate patient reports in line with medical note formats, and structure patient information for EHR
- ethical problems with private clinical data

LLMs for generating synthetic data

- importance of synthetic data for LLMs
 - completely synthetic data used for fine tuning materia can result in models with seemingly better performance
 - e.g. GatorTronGPT creating synthetic medical data, ZeroGen using a pre-trained LLM to create a synthetic movie dataset
 - in chatbot creation, synthetic data is used as a part of instruction-tuning
- existing methods of generating interview transcripts are very time consuming, such as PriMock57: speech-to-text transcripts of mock primary care interviews
- synthetic data generation for chatbots
 - IBM's LAB: Large-Scale Alignment for ChatBots model

LLMs Delivering Semi-Structured Interviews

background conclusions

- current LLMs applied in the medical field are focused on zero-shot or single-shot performance
- synthetic interviews are a need in creating chatbot LLM models

However, we aim to merge the two parts for a pipeline to generate synthetic transcripts. Synthetic interviews may be used for training a smaller model. With both patient and model distinct from each other, different models may be used for each role– for example, a language model that is fine-tuned on more colloquial speech or internet slang or a unique typing style could be implemented. This may result in training sets that can account for more variability in the training data. Training data is important for modifying the outputs of large language models.

In particular, we want to focus on the eventual usage of local large language models.

Methods

Software Architecture

Our pipeline is made up of three main steps: 1. Patient Creation, 2. Transcript Generation, and 3. Transcript Analysis. These steps are separated into their respective directories on the GitHub page. Detailed instructions for setting up the program are available in the README file on GitHub.

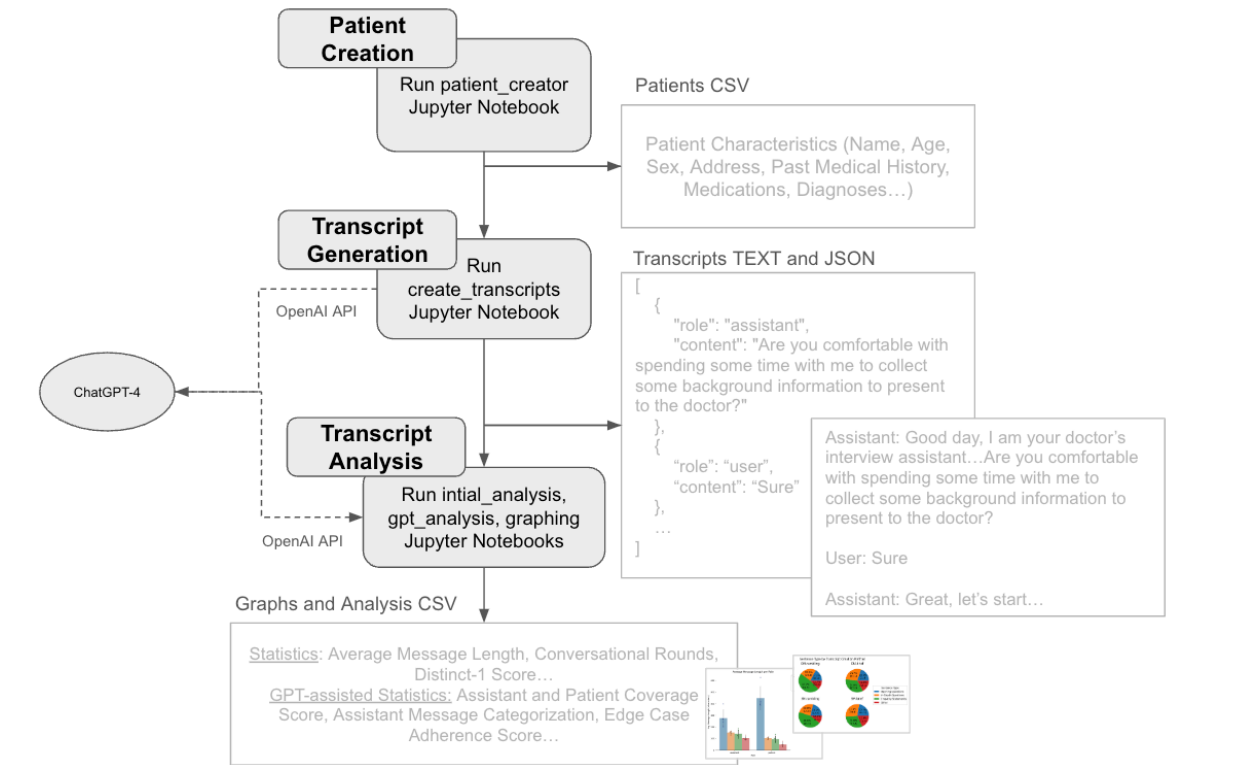


Figure 1. Operational Steps for Synthetic Interview Generation

Patient Creation

Code related to patient creation is in the “patient_creator/” directory.

To add diversity to the synthetic transcripts, we first create a list of synthetic patients with randomized names, ages, occupations, and psychiatric traits. We refer to these randomized elements as patient characteristics. Synthetic patients are stored in a “patients.csv” CSV file, where each row represents a single patient, and each column represents a characteristic. Users can follow steps listed in the “patient_creator.ipynb” Jupyter Notebook to generate multiple patients. This will add a new row to the existing “patients.csv” file.

Some characteristics like name, date of birth, occupation, and address are generated by the Faker Python package **<citation needed>**. Characteristics that cannot be provided by Faker such as medications, common psychiatric conditions, allergies, and typing styles are randomly selected from manually curated variables.

Manually created variables are stored in the “variables.py” Python file. These variables are either lists or dictionaries. Dictionaries are used for dependencies: cases where some characteristics depend on others. For example, someone’s health conditions will limit the medications they are

prescribed. We associate those characteristics together by making a dictionary with the health conditions as keys and a list of prescribed medications as values. When a patient is assigned a health condition, the available medications that can be assigned are constrained to the ones that appear in the list. We created this dictionary by initially asking ChatGPT for the most common psychiatric conditions, and then prompting it to generate the most common medications for treatment. ChatGPT's responses are then manually curated and put into the variables file.

The appearance rate of certain characteristics are also randomized. For example, it is not uncommon for a patient to have comorbid diagnoses, such as anxiety and depression. Thus, we include a chance that one to three psychiatric conditions are given to a patient. Additionally, some characteristics like family of mental health issues are assigned either “yes”, “no”, or blank, for the model to interpret and expand on its own. Patient characteristics, appearance rates and dependencies are assigned in the “patient_creator.py” Python file.

Users can alter the code in “variables.py” and “patient_creator.py” as they see fit. The “patient.csv” CSV file can be manually adjusted after the randomization to include or remove any characteristics. The CSV file is delimited by the “|” character.

A table summarizing the patient generation variables is listed in **Appendix A1**.

Transcript Generation

Code related to transcript generation is in the “transcript_generation/” directory. Double model and Single model transcript generation code is separated into their corresponding directories. Transcript generation prompts are in the “prompts” directory. Users can choose which prompt is used by updating the file path variables in “double_model.py” and “single_model.py” as needed. Finally, users can run “create_transcripts.ipynb” to generate synthetic transcripts for all patients in the “patients.csv” CSV file.

Output synthetic transcripts are output to the “transcripts” directory. Each transcript created follows the naming convention {SM/DM}_{time}{date}_Interview, and is saved as both a TXT file and JSON file. The TXT file is formatted to be human-readable with the tags “Assistant” and “User” for each role's message. The JSON file is arranged in the OpenGPT format, which is a message list of dictionaries with “role” and “content” keys.

Each transcript begins with a standard “starter” message from the assistant, disclosing itself as an LLM-powered chatbot and acknowledging data storage practices. It asks for the patient's agreement to participate in the intake interview process. The starter message is filled in with a randomly generated psychiatrist name and appointment date. An example starter message is provided in **Appendix C**.

Information about the generated transcripts are saved in the “tracking” directory in the “tracking.csv” CSV file. Users can find information about the file name, patient interviewed, time taken, and cost for transcript generation within this file.

GPT API

We interact with the GPT-4 model "gpt-4-0125-preview" using the OpenAI API **<citation needed>**. Users will need to have their own OpenAI API key with access to the "gpt-4-0125-preview" model to use the pipeline. Instructions for setting up the OpenAI API in the program are described in the GitHub README file.

The transcript is created over multiple queries to GPT-4. Each query needs to provide the model name, model temperature, and a message list. The message list is a list of dictionaries, where each dictionary represents a single message sent in the conversation. Each dictionary must have role and content keys. The role value can be either “System” (the system message to the model), “Assistant” (what the model has said), or “User” (what the other party has said). These role keys represent how the chatbot differentiates between messages sent and messages received, rather than representing the “role” that it is being asked to express.

To manipulate how the GPT model will respond, we send specific instructions in the message list with the “System” role. These instructions are referred to as prompts, and tailoring the prompt changes the behavior of the system. In the synthetic transcript generation process, different prompts are used for the Patient and Assistant, as each role has a different objective. The format of our prompts are adapted from Chen et al. (2023)’s previous work on psychiatric large language models as a psychiatric diagnostician or patient.

Question Bank

Transcript generation was created in collaboration with a psychiatrist, who provided a mock interview transcript and an outpatient intake questionnaire. We used these documents as the basis for questions that should be covered in the synthetic interview by extracting topics into a question bank. The final question bank consisted of 47 points with 5 sections titled General Information, Medical History, Family History, Personal History, and Additional Comments. The question bank is used by the assistant to provide structure to its question-asking.

We found that it was necessary to add specific follow-up points in the question bank itself, and provide a final point for ending the interview. The complete question bank is provided in **Appendix C**.

Assistant Prompt Design

The assistant’s task is to gather information about the patient in a semi-structured interview format, covering all the material from the question bank.

The most important aspect of this task is to understand when a patient’s answer is lacking in detail and ask appropriate follow-up questions. For example, if the patient mentioned a previous hospital stay, the assistant should get details for the reason, duration of stay, and how it has affected the patient. If the patient mentions that they have a sibling, the assistant needs to ask about the sibling’s age, name, whether or not they are working. These details are important for the psychiatrist to connect on a more personal level with the patient.

We also want the assistant bot to display empathy. A degree of reflection and acknowledgement during the conversation creates a sense of engagement– an element that questionnaires often lack. Through this sense of engagement, the patient potentially will offer more details about personal experiences and a more comprehensive history. By doing so, the doctor will have more information to consider during the diagnosis stage, and have a richer understanding of the patient’s experiences.

The sections of the assistant prompt are as follows:

1. **Context:** The contextual elements of the assistant’s role is given, including the task it must accomplish.
2. **Question Bank:** The question bank is introduced as “aspects” that the assistant must cover.
3. **Additional Instructions:** We provide the assistant with instructions for further probing the patient. We also provide instructions for handling edge-cases which the assistant must handle. In our example, we ask it to ensure patients report the medication dosages and suggest appropriate responses for typos.
4. **Directions to ask one question at a time:** The fewer the questions, the easier it is for the patient to respond fully. Additionally, this prevents the model from playing a back-and-forth conversation with itself.
5. **Stop Command:** This is only necessary for the double-model generation method. The program monitors the Assistant bot and Patient bot’s back and forth conversation for the presence of a “<STOP>” code, to prevent it from going on an infinite loop. The <STOP> tag is removed from the transcript when it gets turned into a JSON or TXT file. This line is omitted in the single-model generation method as the output can stop on its own.

An example of a filled-out assistant prompt is in **Appendix C**.

Patient Prompt Design

The patient must answer questions based on the patient characteristics given to it. Patient characteristics from the “patients.csv” CSV file are processed into a string and incorporated into the patient prompt.

The patient must also invent plausible events and characters, expanding on details from the provided characteristics. If the characteristics include an edge case scenario, the patient should naturally incorporate it into their response. We found that including a “trigger” for the edge case and including it as the last line helped the model bring it up more consistently.

The sections of the patient prompt are as follows:

1. **Context:** The model is told the task that it is expected to perform.
2. **Patient Characteristics:** The model is told to create a persona with history based on these characteristics, which is generated by the Patient Creator.
3. **Tone directions:** The exact phrasing comes from Chen et al. (2023) which produced results with a higher level of human-like behavior. This involves use of resistance, colloquial language, and emotional fluctuations.
4. **Edge case scenario:** if there is an edge case scenario present in the Patient characteristics, it will be added to the end of the prompt.

An example of a filled-out patient prompt is in **Appendix C**.

Double Model Transcript Generation

For the double model transcript generation method, we create two instances of the GPT-4 model, with one model always receiving the patient prompt (to act as the patient) and the other the assistant prompt (to act as the assistant). For the patient bot, the message list roles must be tagged with “assistant” for its own answers and “user” for the assistant’s questions, and vice-versa for the assistant bot. Otherwise, we found that the back-and-forth aspect breaks down, as both bots become confused in maintaining distinctions between what it has said and what the other entity has said. This devolves into conversations where both bots start to assume the role of the assistant bot and ask questions back and forth.

To operate the double-model method, we append the latest message to the ongoing message list, changing the roles as appropriate. We start off by interacting with the patient model. In the initial message list, we provide the patient prompt as the first “system” message, and the starter message as the second “user” message. Then, the patient model will respond accordingly. We take the patient model’s response and append it to the message list. We then convert the roles in the message list for the assistant by exchanging the first “system” message to the assistant prompt, labeling the second starter message as “assistant”, and the newly appended assistant model’s response as “user”. This loop occurs until the stop condition in the assistant prompt is present in the latest message, indicating that the interview is over.

Single Model Transcript Generation

For the single model transcript generation method, we combine the patient prompt and assistant prompt. The model is prompted to create a transcript that alternates between the two roles. The model must manage both roles simultaneously, switching between being a patient and an assistant. The combined prompt as the assistant's starter message at the end as an example for the model to continue from.

In this case, we send the combined prompt as a "user" message. The model's response should be a complete transcript, with "Assistant:" and "User:" tags to denote the different roles. If the single model generator only responds with the patient's line, the single model generation code should be run again to generate a complete transcript.

Transcript Analysis

Code related to transcript analysis is in the "analysis/" directory.

Firstly, users should run the "initial_analysis.ipynb" Jupyter Notebook to generate objective metrics such as average message length, conversational rounds, and Distinct-1 scores. The output of these metrics will be stored in the "analysis.csv" CSV file.

Secondly, users should run the "gpt_analysis.ipynb" Jupyter Notebook to obtain metrics such as the topic of a given sentence and the completeness of each role's task. We have created an automated manner of extracting quantitative variables by prompting GPT-4 for evaluation. These metrics are stored in the "gpt_analysis.csv" CSV file.

The initial starter message is removed from analyses, because it is an automatic message that is not LLM-generated.

For these analysis methods, we want to measure each role's performance. Thus, we first separate the user's and assistant's messages in the transcript to analyze individually. By performance, we refer to the model's capability to follow the instructions provided for each role.

All graphs were created with code in the "graphing.ipynb" Jupyter Notebook.

Standard Statistics

The length of individual messages sent by the patient can demonstrate adherence to the "Typing Style" characteristic— whether or not the patient successfully follows the "rambling" or "brief" characteristic it was assigned. This is calculated for each transcript by averaging across the length of all patient messages. We calculate these values for the assistant roles too.

Distinct-1 calculates the number of unique words said in a document over the total number of words. We calculate this score for both roles separately. We pass each role’s messages into the Natural Language Processing Toolkit Python Package to receive the numbers of unique words and total words <cite here>.

GPT-Assisted Statistics

Coverage Score

We measure the coverage for both roles by building on an existing prompt template for GPT-assisted evaluation <cite here– Wang et al. 2023?>. Using this prompt, GPT can score LLM-generated text compared to reference text. This prompt allows for flexibility in how the score is determined, such as specifying what aspect of the text should be graded.

We modify the prompt to also include examples for scoring according to our needs by including grading specifications. We also ask ChatGPT-4 to reason through its answer and provide citations to support its answer. This “Chain of Thought” reasoning has been demonstrated to improve the accuracy of LLM responses (Wei et al., 2022). The final prompt templates for patient and assistant coverage are in **Appendix D**.

First, we summarize the patient’s messages in a format that parallels the patient characteristics list that the patient was given. This patient transcript summary is used as the LLM-generated text, and the patient’s original characteristics list is used as the reference text. Then, the summary is compared with the patient information given in the original prompt using the evaluation prompt, and a score out of 100 is given for the coverage rate. Both the patient transcript summary and the evaluation response are saved to the output CSV as the “user_summary” and “user_coverage_content” respectively. The numerical score is extracted and saved in column “user_coverage”.

For the assistant, coverage is evaluated based on a comparison between the question bank topics and the assistant’s messages to receive a score out of 100. The evaluation response is saved to the output CSV in the “asst_coverage_content” column. The numerical score is extracted and saved in column “asst_coverage”.

Assistant Message Categorization

We further evaluate the assistant’s behavior by coding each of its individual sentences. We split up the assistant’s messages into a dictionary, with the key as the message index number and the value as the list of the message’s individual sentences. Sentences are split using the Natural Language Processing Toolkit’s tokenize library and sent_tokenize() function <cite here>.

Allowing GPT-4 to process each sentence individually can prevent miscategorization. A message may contain multiple questions along with empathy statements. For example, consider this assistant message: “Thank you for sharing that with me. Moving on, have you had any previous hospitalizations?” This message contains both a statement and a question. By separating the message into individual sentences, each of these sentences can be categorized accordingly.

For each sentence, we call GPT-4 with a prompt to respond with the most suitable category from the following:

1. “Opening Question”, defined as a question that introduces a new concept.
2. “In-Depth Question”, defined as a question that builds upon a previous question, such as asking for more specific detail or making a clarification.
3. “Empathy Statement”, defined as a statement that expresses reflection, understanding, or empathy.
4. “Other”, which are sentences that do not fall into any of the above.

The full prompt can be found in **Appendix D**.

We use the same terminology from Chen et al. (2023), where they use the counts of “opening questions” and “in-depth questions” to calculate the “in-depth” ratio. The “in-depth” ratio is measured as the number of in-depth questions over the number of total questions. We save the number of opening questions, in-depth questions, empathy statements, and other sentences into the CSV as “asst_opening_qs”, “asst_indepth_qs”, “asst_empathy_statements”, and “asst_other_statements”.

These categories are saved into a dictionary with keys of message index numbers and values of a list with those categories, in order of sentence appearance. This category dictionary is used to calculate the number of questions asked per message, and the GPT-deemed questions can be isolated for statistics to be computed individually. The assistant message dictionaries and category dictionaries are saved to the CSV in columns “asst_sentences_dict” and “asst_category_per_msg” respectively. The number of questions asked per message is saved to the CSV in the column “asst_avg_qs_per_msg”.

Edge Case Adherence

We also want to analyze whether or not the transcript contains the edge case scenario. The edge cases are conversational rounds where the assistant must exercise specific conditions. Examples of this may be if the patient asks a clarifying question, if the patient’s response is vague or not descriptive enough, or if the patient expresses intentions to hurt themselves. functional ability for patients to bring up edge case scenarios and assistants to follow edge case instructions.

We obtain a score by asking GPT-4 to respond with whether or not an edge case behavior is found in a list of messages. In the scenario where the patient makes a typo when spelling their medication, the assistant should make a suggestion of the correct spelling. Thus, on the patient's side, we should find a message that contains a medication name misspelling, and on the assistant's side, we should find a message that contains a medication name correction. These distinctions should be noted in the prompts for each respective role. The prompts used for determining edge case adherence are provided in **Appendix D**.

Edge case adherence can be labeled with one of three outcomes:

1. "Hit" means that the presence of an "edge case" in the prompt matches with the transcript. Either there was an edge case specified and the edge case showed up in the transcript, or the edge case was not specified and there were no such edge cases in the transcript.
2. "Type 1 Error" means that the edge case scenario was included in the prompt, but was not found in the transcript.
3. "Type 2 Error" means that the edge case scenario was not included in the prompt, but was found in the transcript.

Results

Patient Creation

For our study, we created 10 randomized patients by running the "patient_generator.ipynb" Jupyter Notebook. To compare variations in patient response styles, we altered the typing style characteristics so that there was an even distribution of "brief" and "rambling" variations. We created and randomly assigned half of the patients an edge case scenario of misspelling a medication name.

Details about these patient pairs are listed in **Appendix D2**.

Transcript Generation

Each patient profile was used to generate both a single model transcript and a double model transcript, creating a total of 20 transcripts.

The patient model is expected to make up details associated with their assigned categories. When the model is given the prompt "Family History of Health Conditions: yes", it can generate a story about the patient's mom and grandfather:

(Excerpt from SM_20240429-105134_Interview)

Assistant: It sounds like those were very challenging times for you. It's important to have support during such periods. Let's shift a bit and talk about your family. Could you tell me about any psychiatric conditions or neurological or genetic conditions in your family?

User: Yeah, my mom had depression, pretty bad. And my grandpa was diagnosed with Alzheimer's before he passed. It's like a grab bag of fun in my family history.

(Excerpt from DM_20240429-105134_Interview)

Assistant: It sounds like you've built a comprehensive support system with professionals who can address different aspects of your health, both mental and physical. Juggling these appointments and maintaining your health is indeed a significant effort, but it's commendable that you're actively engaged in managing your well-being.

Let's shift a bit and talk about your family history. Do you have any family members with a history of psychiatric conditions? If yes, who, and did they receive treatment or hospital care?

User: Oh, family history, that's a bit of a tangled web. Yeah, there's some of that running through the family. My mom, she had her own battles, you know? Depression, I think, though it was never really talked about much. She had a hard time of it, especially after my dad left. I don't know all the details, but I remember there were days she just couldn't get out of bed, and then she'd be okay for a while, up and doing things, almost like nothing was wrong. But she never went to a hospital or anything. Back then, it wasn't something people did, or if they did, they didn't talk about it. And my uncle, on my dad's side, he was in and out of places when I was really little. I think it was schizophrenia or something serious like that. So yeah, there's a history. Makes you wonder how much of this stuff is passed down, you know?

These two excerpts are created from the same patient profile, which contains the “rambling” characteristic. This is one example from many where it is apparent that the double model transcript generation method does appear more “rambly” than the transcript from the single model method.

A patient model with patient profile with the characteristics “Medical Conditions: attention-deficit/hyperactivity disorder, borderline personality disorder, schizophrenia” and “Developmental Difficulties: yes” resulted in a storyline consistent between its medical conditions and childhood experiences:

(Excerpt from DM_20240428-222418_Interview)

Assistant: Good to hear it was straightforward. At a young age, did you walk, talk, and develop friendships like other kids, or were there any notable differences?

User: Yeah, I guess I was a bit slower on some of those fronts. Had trouble keeping friends, and they said I was always a bit more in my own world. ADHD diagnosis came because I couldn't sit still or focus much.

Assistant: Thank you for sharing that insight. It helps to understand the context of your ADHD diagnosis. Did you face any difficulties during your elementary school years?

User: Was average, I suppose. Struggled to pay attention, got into trouble for daydreaming or not following instructions. Teachers always said I could do better if I just focused more. Classic ADHD stuff, really.

There are also demonstrations of edge case navigation abilities from both the patient and assistant, where the patient was instructed to misspell their medication (patient profile had the characteristics “Medications: Clonazepam” and “Edge Case Scenario: Upon first asking about medication, please misspell the name of the medication you take. You do not need to correct yourself.”):

(Excerpt from DM_20240428-221529_Interview)

Assistant: Noted, I'll make sure Dr. Phillips is aware of your allergies. Could you tell me about any current medications and their dosages that you're taking?

User: Taking Clonazepam for the panic. Sorry, not great with spelling on that one.

Assistant: That's alright, you're likely referring to Clonazepam. Do you know the dosage you're currently taking?

User: Yeah, it's 0.5 mg. Take it as needed, not every day.

(Excerpt from SM_20240428-221529_Interview)

Assistant: It's important to have that noted. What about current medications and their dosage?

User: I take Clonazepam for my nerves. But I'm not sure if I spelled that right.

Assistant: That's okay, Deb. You're likely referring to Clonazepam. Do you know the dosage you're currently taking?

User: It's 0.5 mg, I take it twice a day.

And there are also cases where the patient misspells their medication but the assistant does not catch it:

(Excerpt from DM_20240428-222702_Interview)

Assistant: Definitely, it's good to have one less thing to manage. Could you tell me about any current medications and their dosages that you are taking?

User: Oh, yeah, I'm on this med for my mood, called Fluxetine or something. I'm terrible with names. The dosage... I think it's like 20 mg? Yeah, that sounds about right. Not sure it's doing much, but they tell me it's supposed to help.

Assistant: Fluoxetine, understood. It's quite common for managing mood. Are you taking any health supplements?

Synthetic transcript Analysis for single and double patient models, n=10 patients.

Measure	DM-rambling	DM-brief	SM-Rambling	SM-brief
Assistant Average Message Length (characters)	275.810708	151.945341	141.615278	102.691147
Patient Average Message Length (characters)	447.859871	103.218603	95.508709	48.263711
Average Conversational Rounds	37.6	39.4	23.9	29.6
Assistant Distinct-1 Score	0.287959	0.308299	0.360970	0.379467
Patient Distinct-1 Score	0.203061	0.356550	0.409475	0.479860
Assistant Coverage Score	95.800000	96.200000	55.800000	66.000000
Patient	96.00000	91.000000	88.200000	42.20000

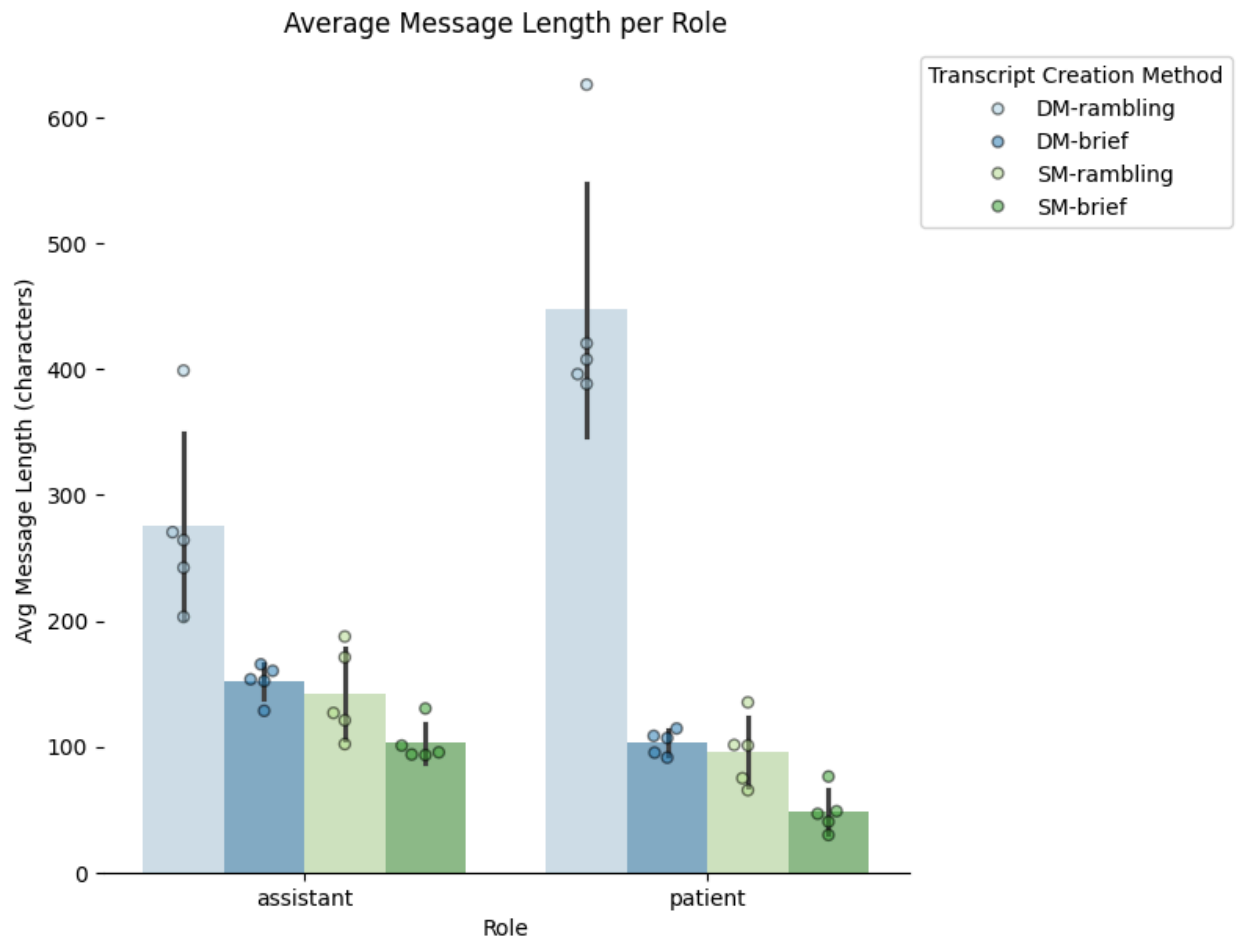
Coverage Score				
Assistant Empathy Statements Ratio	0.473835	0.346911	0.449426	0.296775
Assistant In-Depth Ratio	0.585779	0.580557	0.518208	0.553035
Assistant Average Num Questions per Message	1.461784	1.405299	1.228626	1.171270

Table 1. Mean Measures by Transcript Generation Method

Standard Statistics

Transcripts with the same transcript creation methods are grouped together: “DM” or “SM” denotes “double model” or “single model” respectively, “rambling” or “brief” denotes that those characteristics were in the patient’s prompt.

Average Message Length



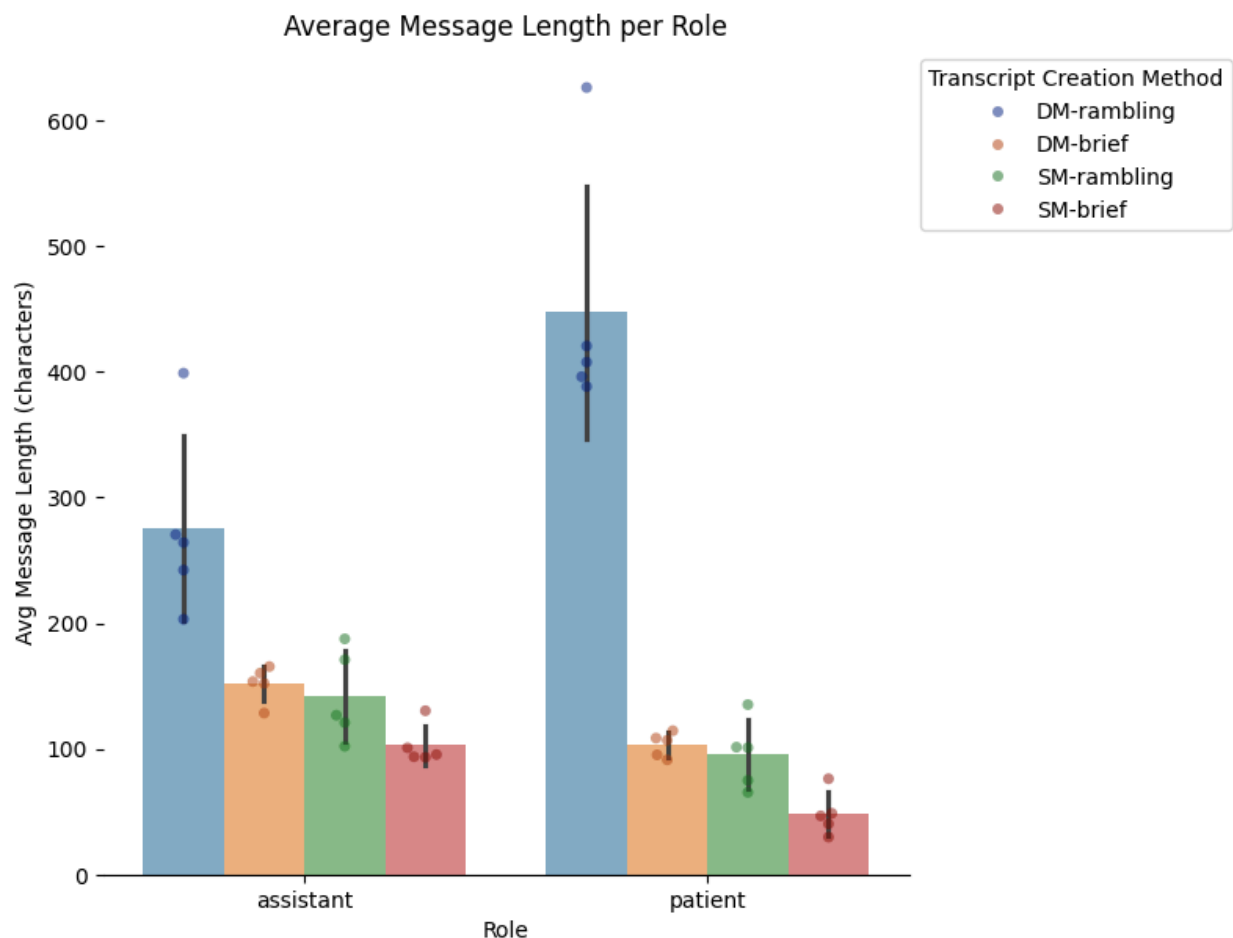


Figure 1. Average Message Length by Role

The average message length is the average character count across each role's messages.

There are significant differences between the patients' average message lengths for DM-rambling and DM-brief ($p=0.0000612$), DM-rambling and SM-rambling ($p=0.0000651$), DM-rambling and SM-brief ($p=0.0000225$), and DM-brief and SM-brief ($p=0.000255$). There are significant differences between the assistants' average message lengths for DM-rambling and SM-brief ($p=0.000891$), and DM-brief and SM-brief ($p=0.000823$). Significance is calculated by t-tests with Bonferroni correction for multiple comparisons ($\alpha = 0.05/6 \text{ comparisons} = 0.00833$).

Conversational Rounds

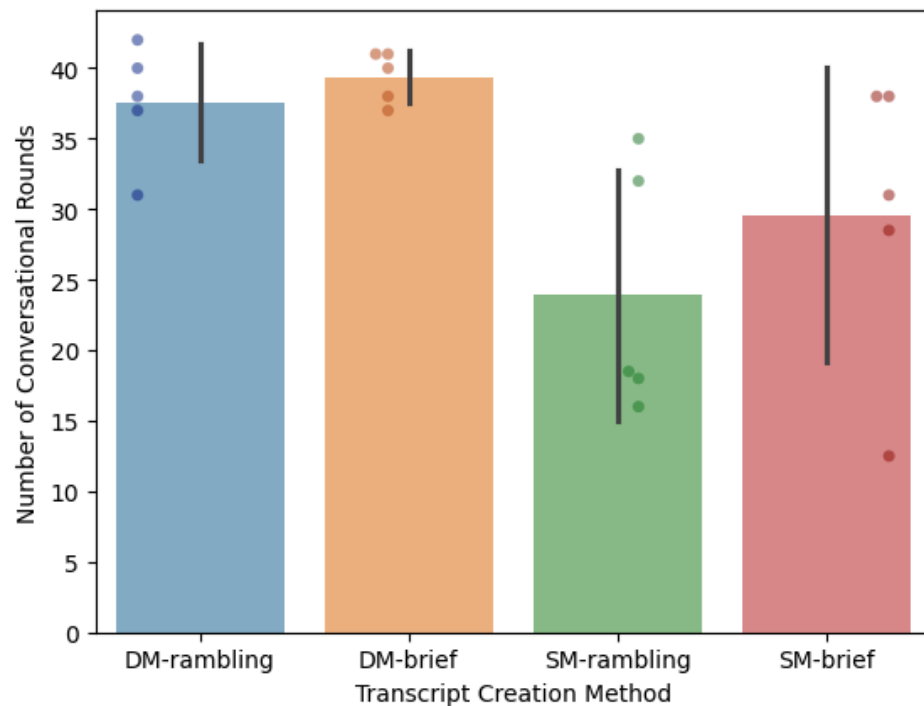
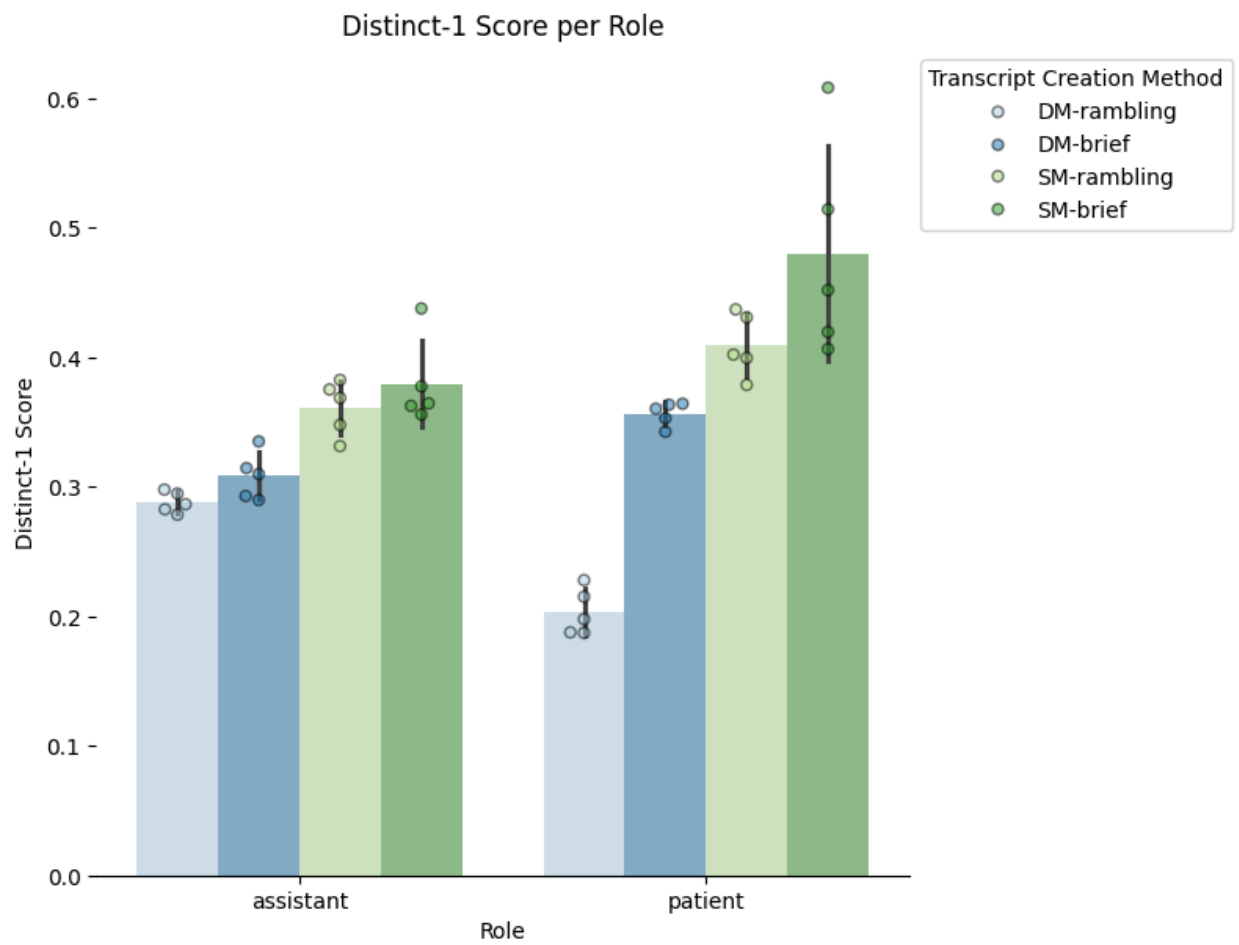


Figure 2. Conversational Rounds by Transcript Creation Method

Each conversational round is defined as one set of back-and-forth conversation, with the user’s message first and the assistant’s message afterward. We use this layout to be consistent with the terminology used in LLM training, where the user’s query prompts a response from the LLM. There are no significant differences between transcript creation methods.

Distinct-1 Score



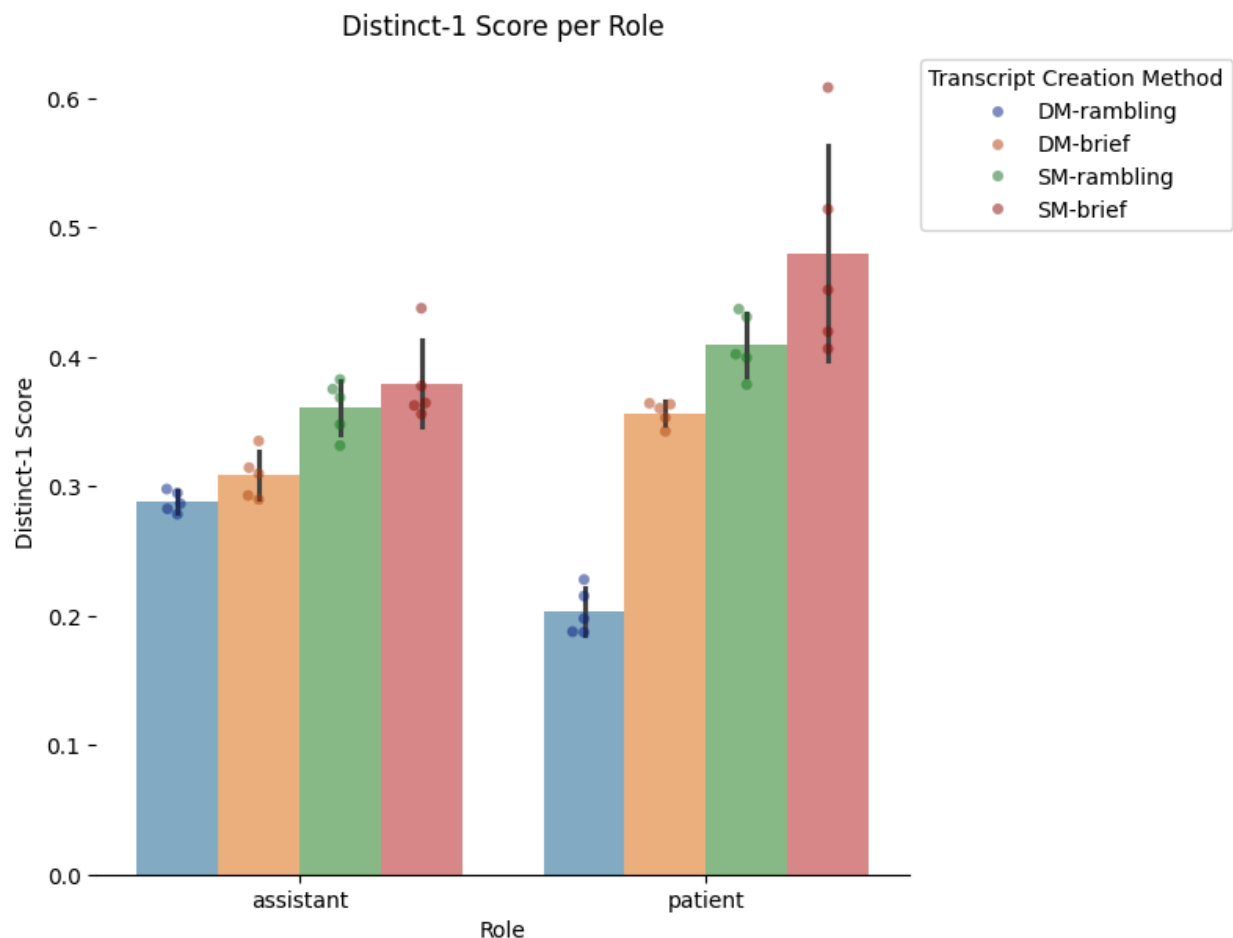


Figure 3. Distinct-1 Score per Role

Distinct-1 Score measures the number of unique words over the total number of words present in text. Chen et al. (2023) suggests that a lower Distinct-1 score implies more colloquial speech, which is more preferable for the assistant, as users may not always understand technical language.

Between patients, significant differences were found between DM-rambling and DM-brief ($p=0.000000138$), DM-rambling and SM-rambling ($p=0.000000315$), DM-rambling and SM-brief ($p=0.0000837$), and DM-brief and SM-rambling ($p=0.00175$). Between assistants, significant differences were found between DM-rambling and SM-rambling ($p=0.0000892$), DM-rambling and SM-brief ($p=0.000340$), DM-brief and SM-rambling ($p=0.00288$), and DM-brief and SM-brief ($p=0.00306$). Significance is calculated by t-tests with Bonferroni correction for multiple comparisons ($\alpha = 0.05/6 \text{ comparisons} = 0.00833$).

GPT-Assisted Statistics

Coverage Score

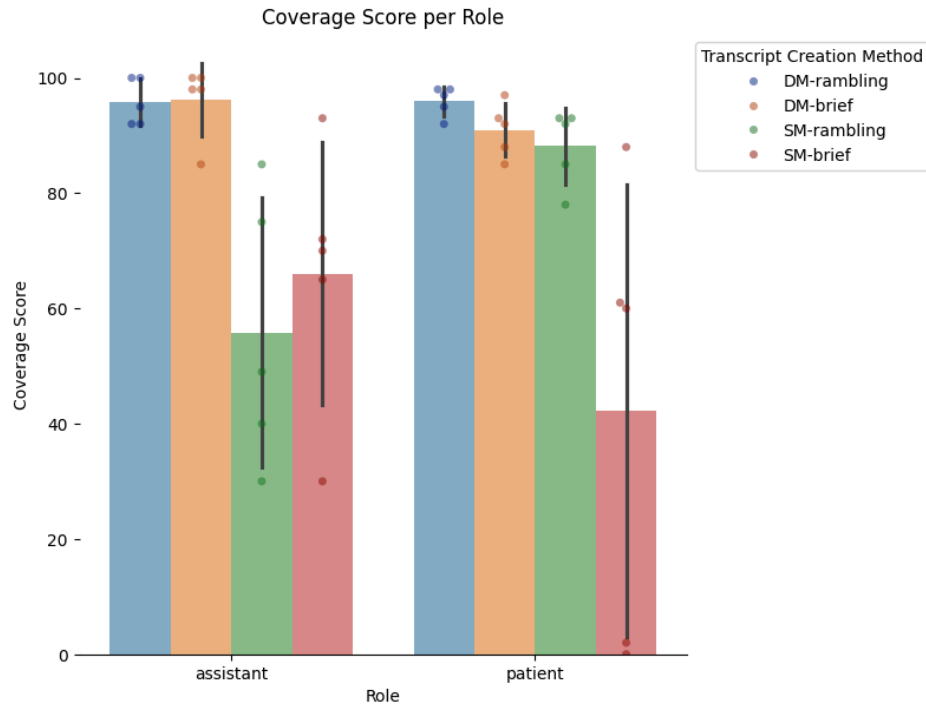


Figure 4a. Coverage Score with Typing Style Distinction

The coverage score for the assistant is determined by the coverage of the assistant's messages and the question bank in its prompt. The coverage score for the patient is determined by the coverage of a summary and the patient characteristics list in its prompt. There were no significant differences found between the means of transcript creation methods.

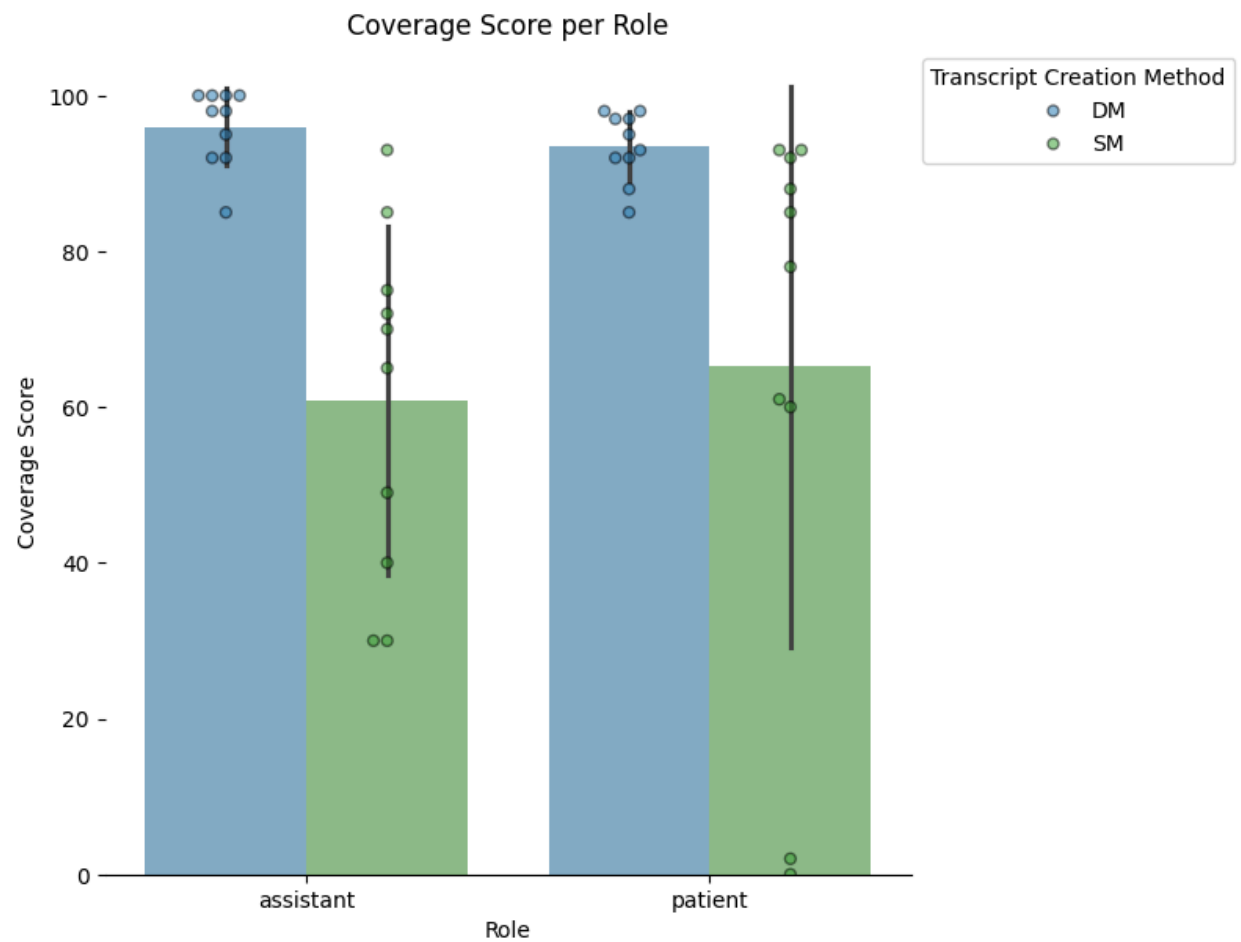
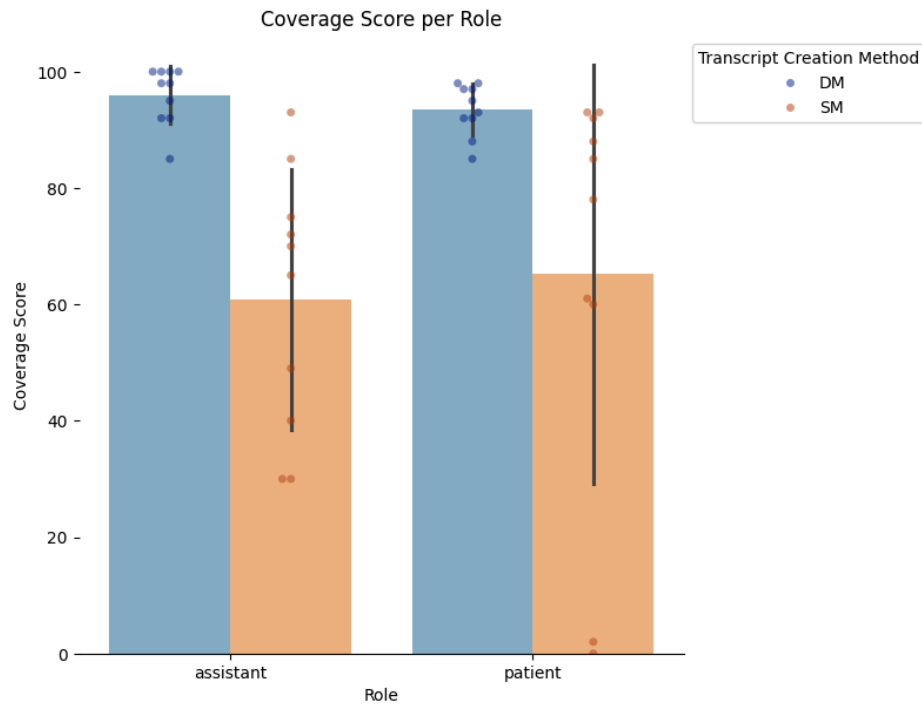


Figure 4b. Typing Style without Typing Style Distinction

Significant differences were found between the transcript creation methods after the typing style grouping is taken away. (Double model assistant: mean=96.000, SD=5.011; Double model patient: mean=93.500, SD=4.403; Single model assistant: mean=60.900, SD = 22.412; Single model patient: mean=65.200, SD=35.946)

The Coverage Score of the assistants are significantly different between the means of double-model and single-model generation methods by t-test ($p=0.000133$).

Empathy Statements

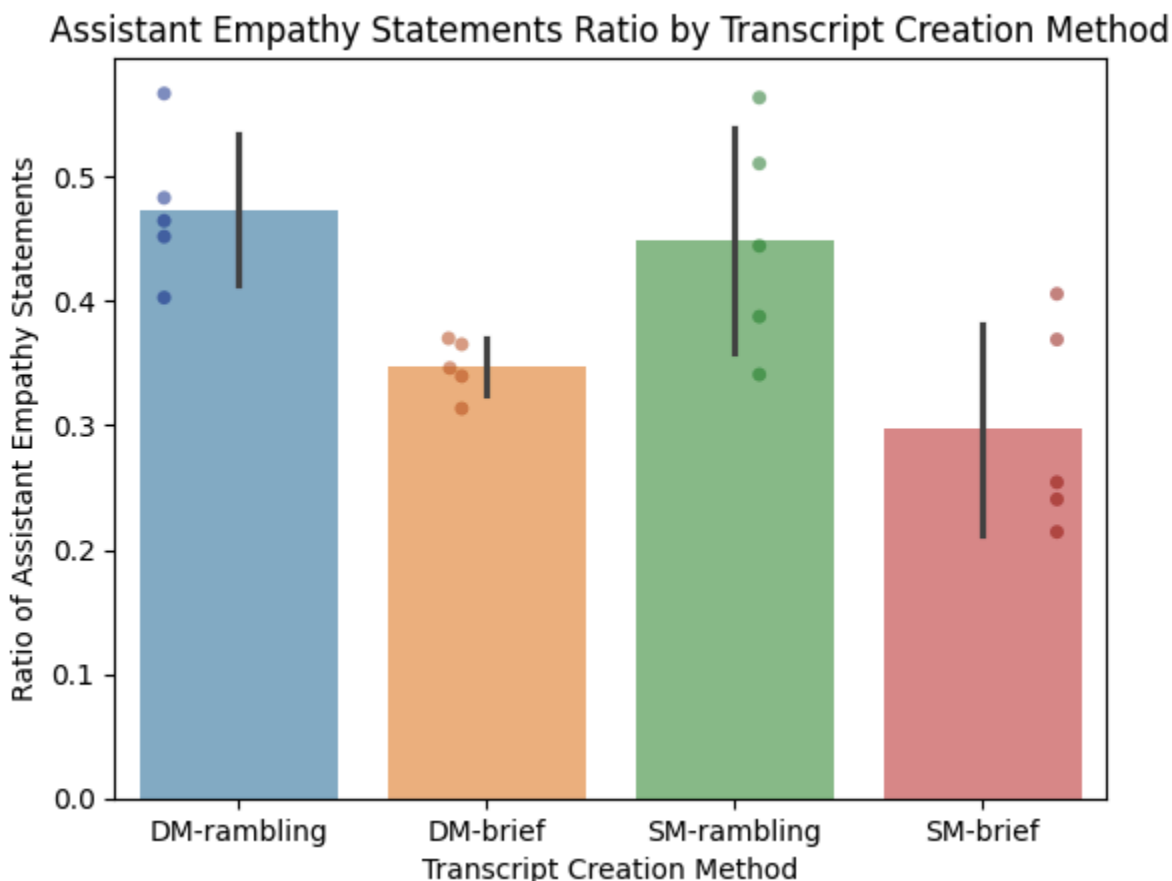


Figure 5a. Assistant Empathy Statements by Transcript Creation Method

Empathy Statements Ratio reflects the number of empathy statements in the transcript over all assistant sentences. We want the assistant to be empathetic to the patient's responses to foster a sense of trust and connection. Significant differences were found between DM-rambling and DM-brief ($p=0.0000413$), DM-rambling and SM-rambling ($p=0.0000713$), and DM-rambling and SM-brief ($p=0.0000482$). Significance was calculated by t-tests with Bonferroni correction for multiple comparisons ($\alpha = 0.05/6 \text{ comparisons} = 0.00833$).

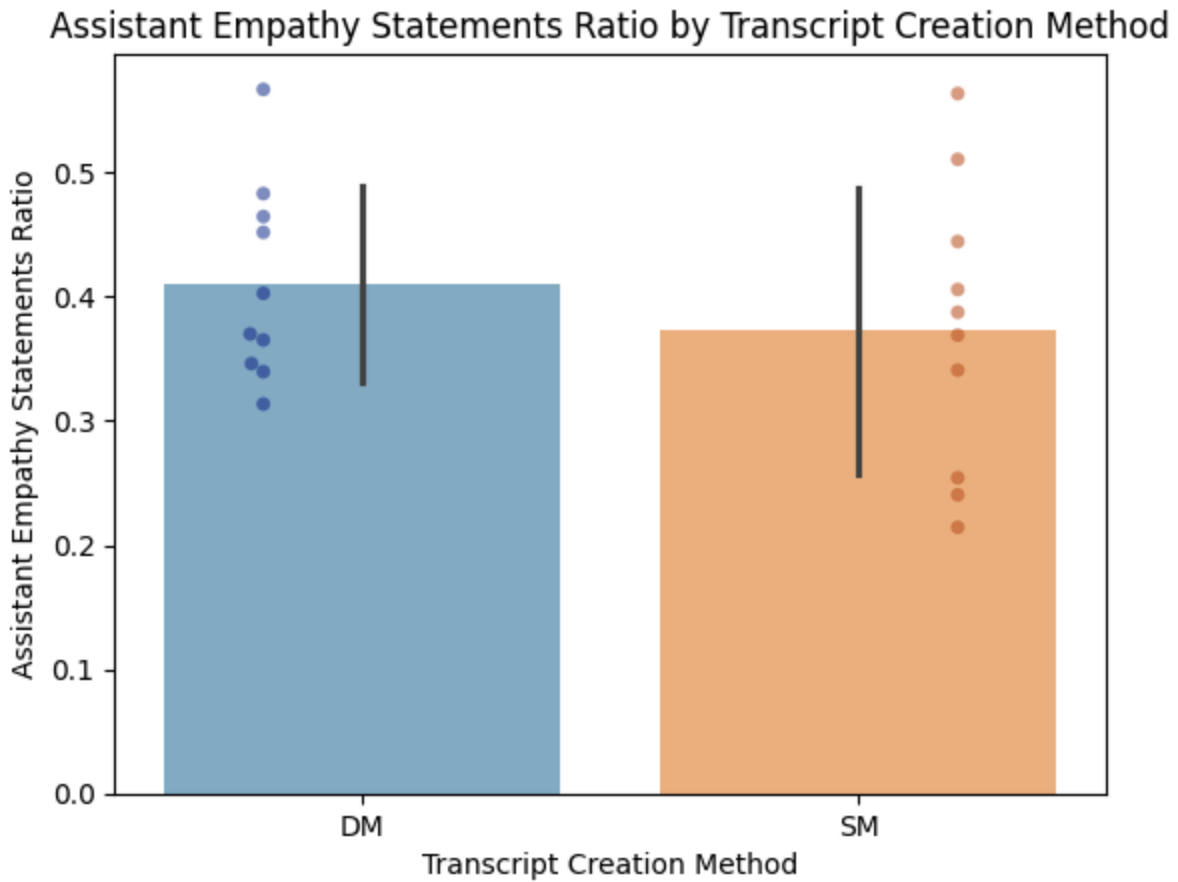


Figure 5b. Assistant Empathy Statements by Transcript Creation Method without Typing Style Distinction

Without the typing style distinction in the patient’s prompt, there were no significant differences found between double-model and single-model generation methods.

Average Questions per Message

Assistant Average Questions per Message by Transcript Creation Method

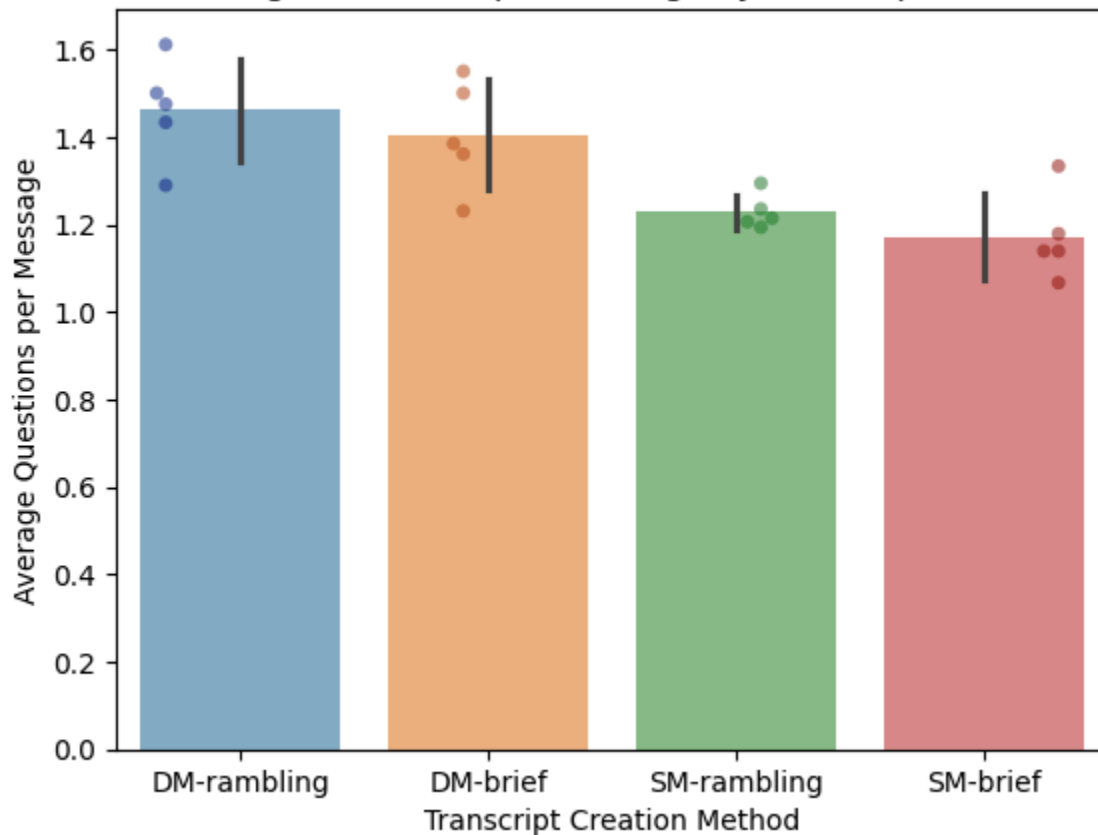


Figure 6a. Average Questions per Message

Average questions per message refers to the number of questions asked for each separate assistant message. Fewer questions asked per message is more desirable; asking one question at a time is most user-friendly, since asking multiple questions leaves room for accidental omissions by the user (Chen et al., 2023). There were significant differences found between DM-rambling and SM-rambling ($p=0.00287$), and DM-rambling and SM-brief ($p=0.00282$). Significance was calculated by t-tests with Bonferroni correction for multiple comparisons ($\alpha = 0.05/6$ comparisons = 0.00833).

Assistant Average Questions per Message by Transcript Creation Method

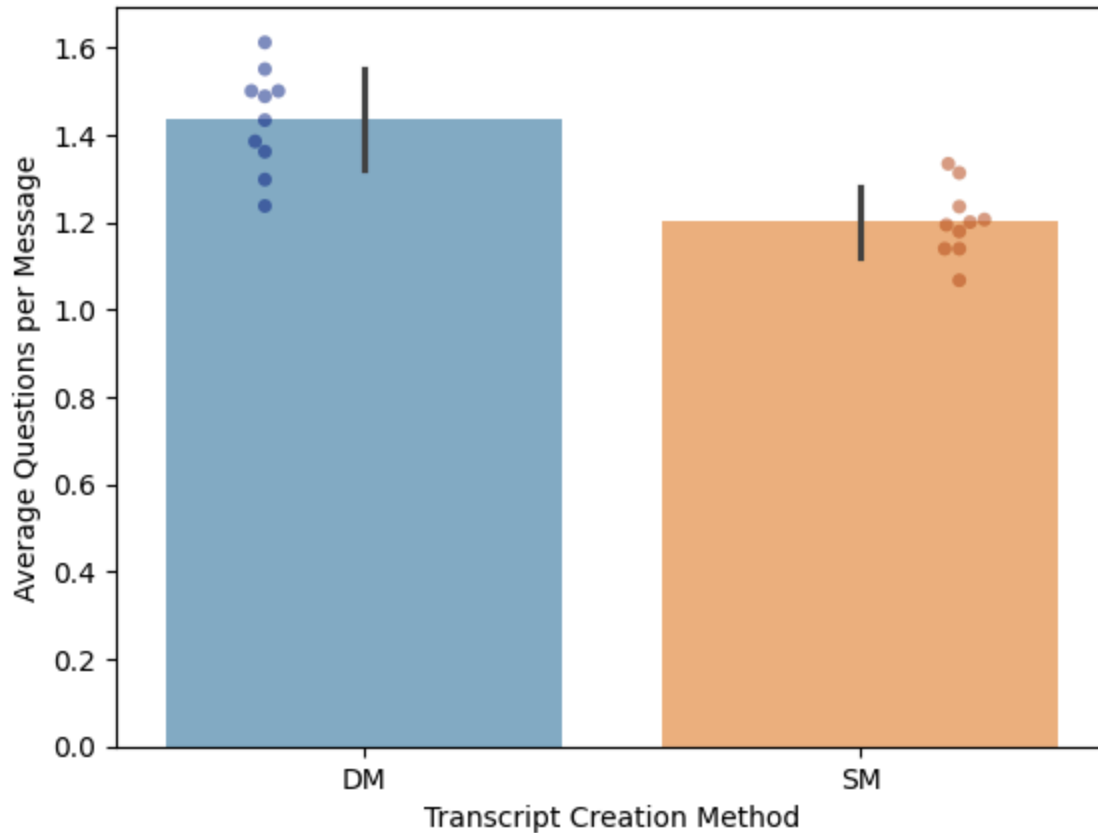


Figure 6b. Average Questions per Message without Typing Style Distinction

Without the typing style distinction in the patient's prompt, there were significant differences found between double-model and single-model generation methods ($p=0.0000501$). Significance was calculated by t-tests ($\alpha = 0.05$).

Assistant In-Depth Ratio

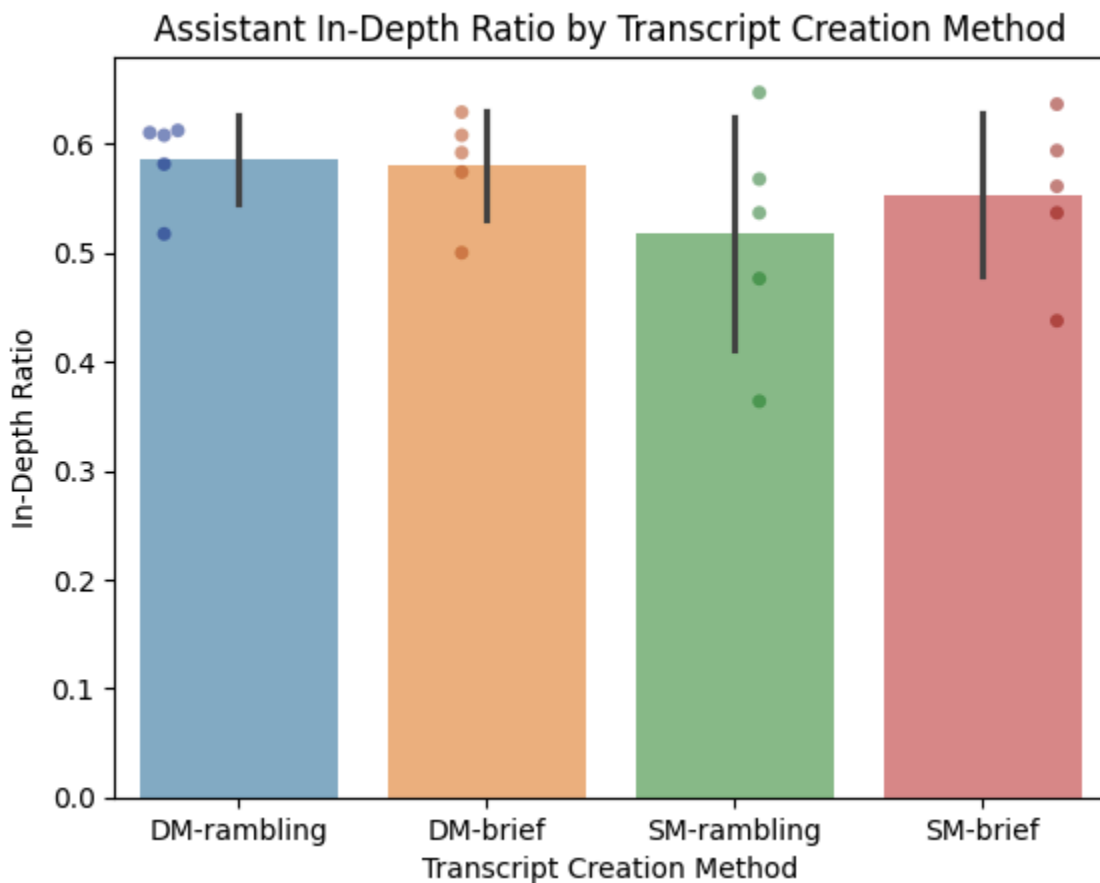


Figure 7a. Assistant In-Depth Ratio

The in-depth ratio for the assistant is the number of in-depth questions over the number of total questions. This is based on existing work done by Chen et al. (2023), which looks at the same metrics. There were no significant differences found between any generation method pairings.

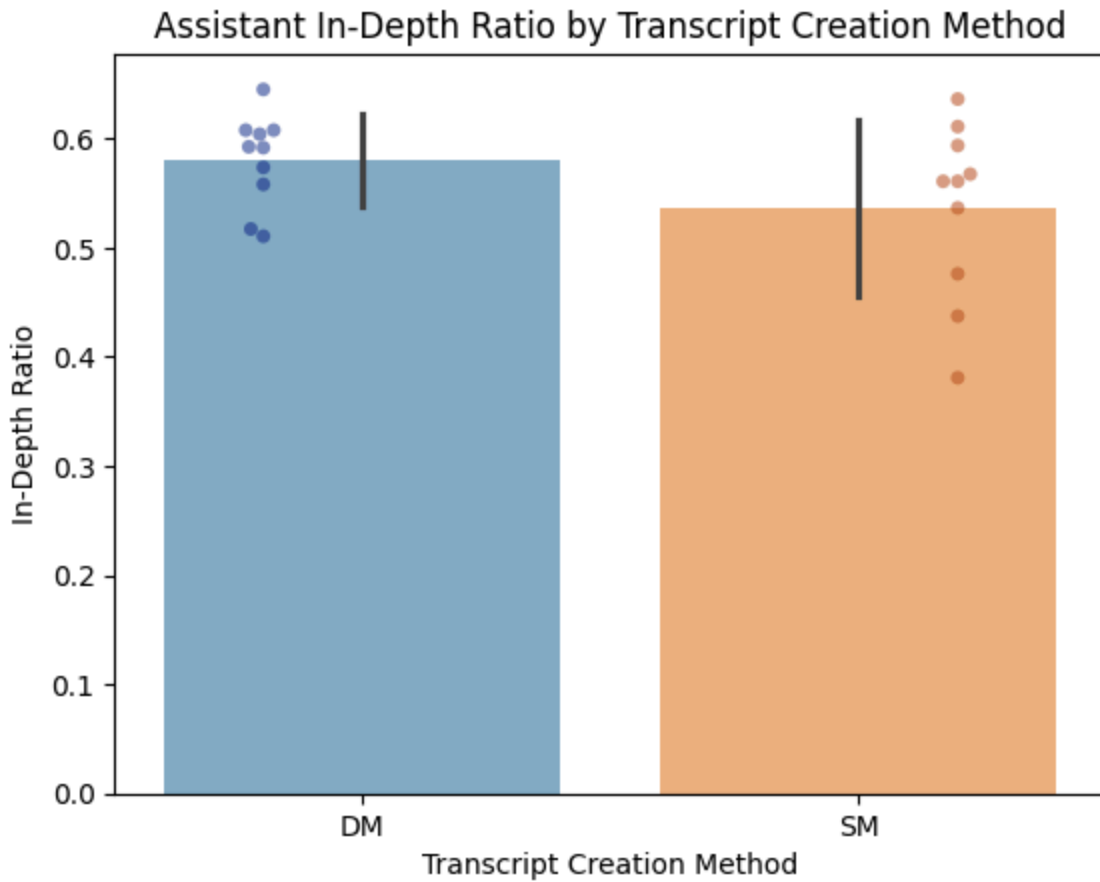


Figure 7b. Assistant In-Depth Ratio without Typing Style Distinction

The in-depth ratio for the assistant is the number of in-depth questions over the number of total questions. No significant differences were found between the transcript generation methods when the typing style distinction is removed.

Edge Case Adherence by Transcript Creation Method

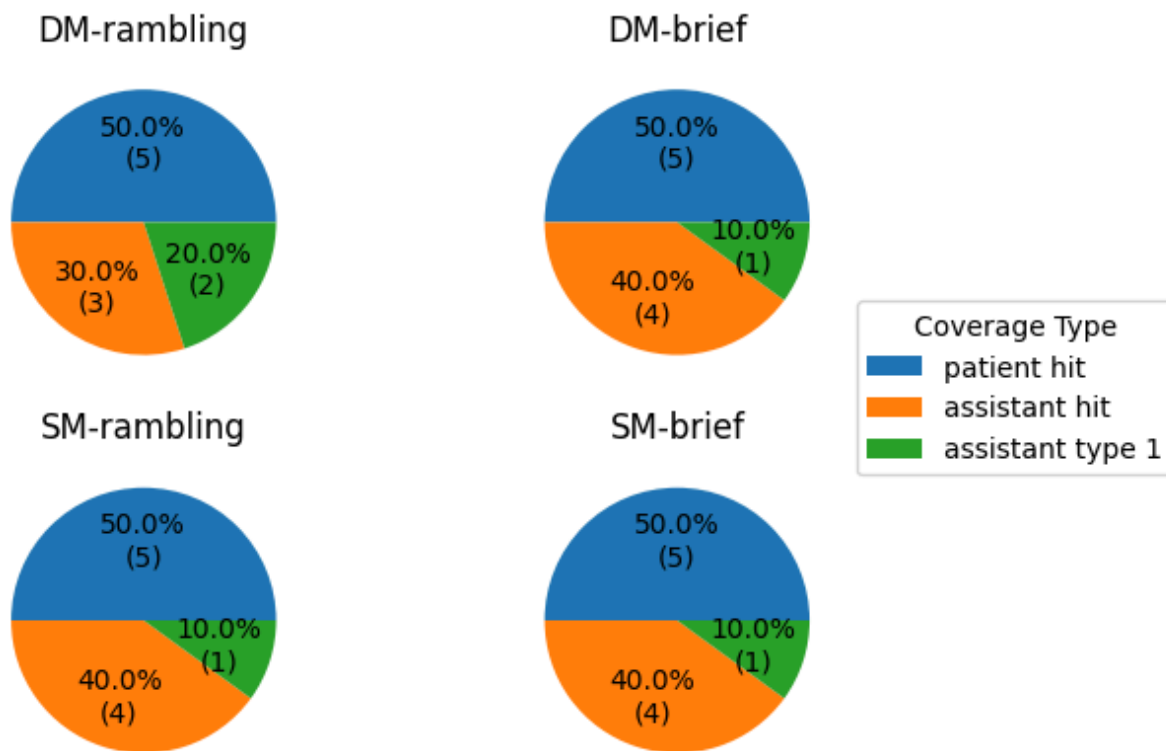


Figure 8. Edge Case Adherence

Coverage type refers to how the edge case has been handled according to the input prompts:

- “Hit” means that the presence of an “edge case” in the prompt matches with the transcript.
- “Type 1 Error” means that the edge case scenario was included in the prompt, but was not found in the transcript.
- “Type 2 Error” means that the edge case scenario was not included in the prompt, but was found in the transcript.

The edge case in the synthetic transcripts is “Upon first asking about medication, please misspell the name of the medication you take. You do not need to correct yourself.” In this pie chart, assistants and patients are assigned different colors for differentiation.

shows whether the edge case is executed in the transcript or not. The three scenarios are as follows:

“hit” meaning the edge case is in the transcript and the prompt, or vice versa

“false negative” meaning the edge case is not in the transcript is in the prompt

“false positive” meaning the edge case is not in the transcript but is in the prompt

Sentence Type by Transcript Creation Method

Sentence Type by Transcript Creation Method

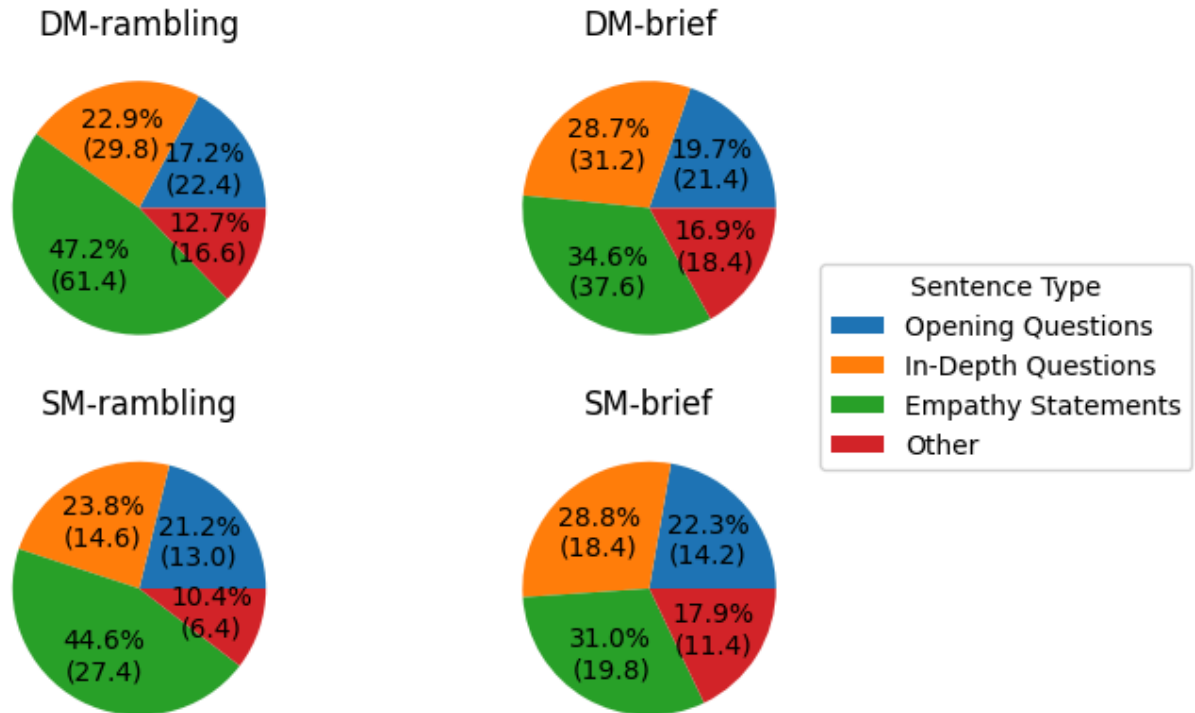


Figure 9. Sentence Type Pie Chart

Average number of sentence types for each transcript generation method.

Question Type by Transcript Creation Method

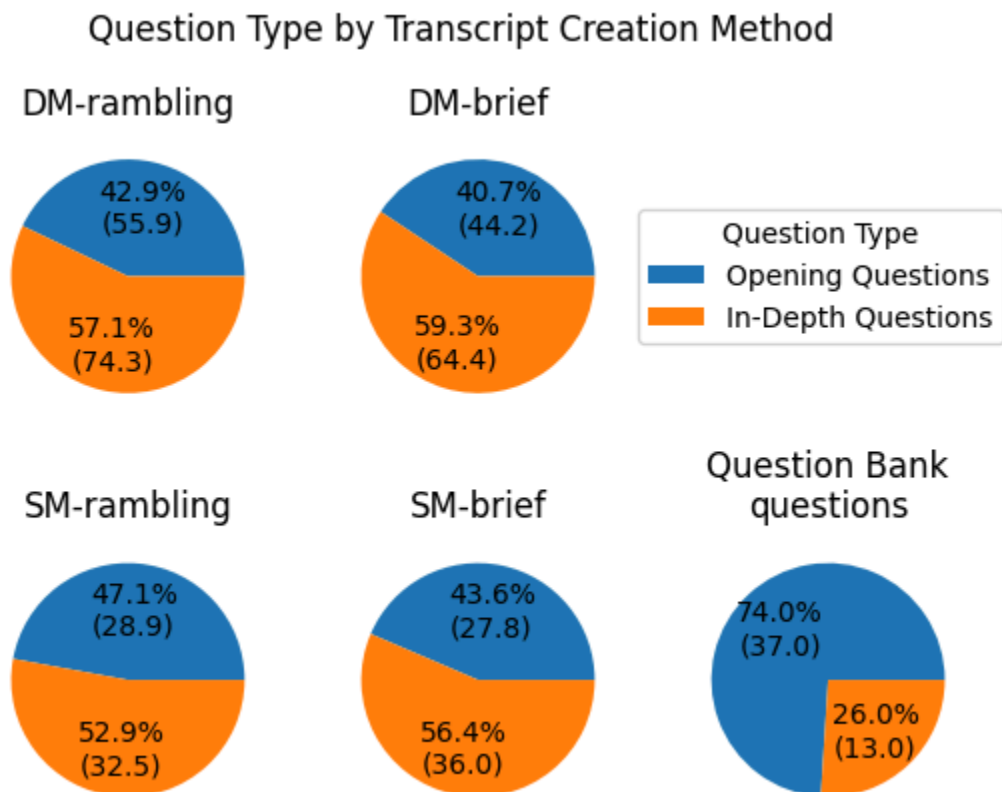


Figure 10. Sentence Type Pie Chart

The average number of opening and in-depth questions asked per transcript creation method. The original distribution of opening and in-depth topics listed in the question bank is shown for comparison.

Discussion

Patient Creation

We create synthetic patients for three reasons. Firstly, we found that generating “random” characteristics from GPT alone was not diverse enough. The model will often respond with the same occupations such as “designer” and “computer engineer” per query. This may be due to how these occupations may be overrepresented in the training data, and have the highest probability of being chosen. Secondly, we can alter variables that we want the patient to express by exporting the patients methodologically to a CSV. In our demonstration, we chose to set the typing style as either “rambling” or “brief” to see how this alters the patient model’s performance, as other variables are held constant. Thirdly, the creation of this CSV and the generation method allows for us to track the degree of representation across the patient personas created.

Certain characteristics are generated but not used in the patient prompt, such as “Reason for Appointment” and “Conversational Tone”. This was implemented at first to create more varieties in the patient bot’s tone and content, but were difficult to measure or decreased overall performance.

Further tuning of the variables list may be necessary for generating specialized patients. It would also be beneficial to further ground these variables in real life by drawing it from existing population data. The patients created by the “patient_generator.py” program have fake addresses in fake cities in British Columbia, Canada, as this is the population where the deployed assistant is expected to serve. However, a drawback of the Faker package comes from the lack of diverse cultural names— though British Columbia and Canada as a whole have significant immigrant populations, the Faker package tends to generate common English first names. It would be best for the characteristic appearance rates to be optimized for a realistic distribution that represents the population that the model will interact with.

Transcript Generation and Analysis

Transcripts created by the double model and single model generation methods differ greatly on the possible message lengths that each patient can output. By varying the “Typing Style” characteristic in the patient prompt, we do see significant differences between the lengths of messages that are consistent with the prompt. The patient in the double model generation method with the “rambling” characteristic results in significantly higher patient message lengths compared to all other transcript methods. The average mean assistant message length generated by the double model method and “Rambling” typing style is 275.810, with a maximum of 626.354 average characters. Meanwhile, the average mean assistant message length generated by the single model method is 95.508, with a maximum of 135.166 average characters (Table 1, Figure 1).

This may be due to context size limitations for the GPT-4 model. Context size refers to the maximum number of tokens that LLMs can work with. The number of tokens in an entire prompt is calculated as the number of tokens in the LLM input (the system prompts and content in message list) added to the number of tokens in the LLM output (the response from GPT-4). This is a natural limitation dictated by the LLM model’s size and available memory capacities. Because LLMs output words consecutively, there is a certain amount of tokens it can work with. It could be the case that single-model transcripts must be created in one GPT-4 output message, the context size is greatly reduced. Context size is relevant in how much a model is able to pay attention to the previous tokens to generate the next token. When the prompts in the system message are further away from the end of the output, the LLM may stop paying attention to its initial directions. This results in an inherent limiting factor.

This limiting factor also resulted in a drop in assistant and patient coverage scores between double model (average assistant score = 96.000, average patient score = 93.500) and single model (average assistant score = 65.200, average patient score = 60.900) generation methods (Figure 4b). There is a significant difference between the average score of the two roles, but it could also be in part due to abrupt stops in a few of the single method transcripts (SM_20240428-215539_Interview, SM_20240428-220920_Interview, and SM_20240428-222418_Interview.txt). As the single model does not consistently output a full transcript even when the same prompt is queried again, this may point towards a need for further prompt engineering. It may also be viable to implement automated incompleteness detection in the code itself.

The double model method appears to be more consistent in its coverage even across longer transcripts with larger message list sizes. This could mean that OpenAI implements its own methods to maximize GPT-4's context size, by keeping the system message consistent and either implementing a moving window to limit the number of messages that the model can see at one time <cite here>, or summarizing earlier portions of the message list automatically <cite here>.

Distinct-1 Score

The assistant should speak in more colloquial terms compared to technical terms as to not overwhelm or confuse the patient. A lower Distinct-1 is desirable for both assistant and patient, as a lower score indicates more colloquial speech (Chen et al., 2023). The patients and assistants in the double model generation method both have significantly lower D-1 scores compared to their single model counterparts, with the double model patient and “rambling” characteristic having the lowest of them all (Figure 3).

Average Questions per Message

The lower the average number of questions per message, the better the user experience (Chen et al., 2023). When more than one question is asked in a single message, the greater the mental exertion is required for the user— this can lead to incomplete responses or neglected answers. We want the assistant to ask certain questions and follow-up only when necessary, but the assistant tends to ask follow-up questions present in the question bank in the same message. This can be seen in the average questions per message graph (Figure 6a and 6b), which show significant differences between double model and single model pairings.

The assistant tends to group follow-up questions with the main question, regardless of whether or not the main question applies to the patient: “That's good to hear, Oscar. It helps to narrow down the areas we need to focus on. Have you ever had any history of rehabilitation or substance counseling? If yes, could you share where and when?” (Excerpt from DM_20240428-215539). This means it may be necessary to further adjust the assistant prompt, since the assistant overlooks the part in the prompt to “ask one question at a time”.

Sentence and Question Types

Certain sentence types are overrepresented in the automated GPT-4 categorization. “Empathy statements” are slightly underrepresented, as a few “empathy statements” are sorted into the “other” category. This is due to how, while some sentences do not overtly demonstrate empathy, they build understanding and trust by reflecting the patient’s words. For example, “It helps to have a picture of your family structure” (Excerpt from DM_20240428-221529_Interview) could be noted as an empathetic statement, but the prompt does not categorize it accordingly. More examples of empathy statements could be provided within the analysis prompt.

The in-depth ratio is higher for every transcript creation method compared to the question bank (Figure 10), which suggests that the assistant asks more follow-up questions than it is prompted to. While this could make sense,

Additionally, it is possible that the automated evaluation prompt wrongly categorizes questions as “in-depth” when it should be an “opening” question. For example, for the same DM_20240428-221529_Interview transcript, 16 out of 23 questions labeled “in-depth” should actually be considered “opening”. This means that in-depth questions may be overrepresented in the type categorization process, which results in a higher in-depth ratio than expected. This is due to the specific nature of questions in our question bank and our assistant’s goal. We are borrowing prompt-engineering terminology from Chen et al. (2023). Their assistant model’s goal is to gather patient symptoms (For Chen et al., 2023). On the contrary, our assistant model’s goal is to gather patient history details. “What is your history of seizures?” is an “opening” question in the context of patient history taking, but can be interpreted as an “in-depth” question to the nature or manifestation of seizures. Specific phrasing in LLM prompt engineering is very important, and “opening” and “in-depth” may be terms that contain more

To extract more balanced results, we should include the question bank as an example in the prompt to describe the “open” and “in-depth” question categories that we are looking for. We can implement chain-of-thought techniques here to first sort the sentences as either “question” or “statement”, then continue sorting into the finer “open” or “in-depth” question categories.

Edge Case Adherence

Lastly, the edge case adherence scores were “hits” for the patients in all roles and typing styles (Figure 8). Edge case follow-up was not carried out by the assistant in 5 transcripts, even when the edge case was present (DM_20240428-222702_Interview, DM_20240428-221850_Interview, DM_20240428-222418_Interview, SM_20240428-221850_Interview, and SM_20240428-222418_Interview.json). This means that these errors are classified as “type 1” errors—the assistant is not following through with the edge case scenario even when it is expected of them to do so. Since both double model and single model transcripts show this error, this could mean that the assistant prompt could use more

fine-tuning for the edge case handling direction to be more prominent. Additionally, the error seems to be associated with certain patient profiles, indicating that the model might have difficulties with following up those medication misspellings in particular (Ritalin and Zoloft for -222418 and -221850 respectively). Either transcripts should include more patient profiles should be created with these medication characteristics. Alternatively, those misspellings could be brought up as examples in the assistant's prompt itself.

However, it is interesting to note the 100% accuracy rate for all patients in carrying out the assigned edge case scenario, regardless of transcript generation type. This method and prompt could be viable templates for future work in training dataset synthesis and creation.

Limitations and Next Steps

If synthetic transcripts must be used for training assistant LLMs, the most important marker of training success is in real-world scenarios. The only way to measure this would be to fine-tune or train an LLM with the synthetic transcripts, and test the model's performance with human users. The pipeline we created provides helper methods for turning text transcripts into JSON-formatted files, formatting the JSON files into message rounds for LLM training, and analyzing transcripts in an automated fashion. While we provide examples of completely synthetic transcripts, these techniques can also be used for transcripts collected with human collaboration.

Further improvements to this pipeline would be to firstly identify the prevalence and types of LLM edge case concerns to incorporate into the edge case scenarios. The assistant may need to answer a patient's clarifying answer, or calm down a frustrated patient. It may also be interesting to test for the patient's ability to bring up multiple edge case scenarios, as more than one may arise in a transcript.

Secondly, it could be interesting to explore how fine-tuned models may be substituted for the patient and user models. We had difficulties getting the GPT-4 patient to incorporate edge case scenarios with mentions of suicidality, or incorporate misspellings with irregular grammar in a consistent manner. It could be interesting to substitute the GPT-4 patient with another model that does not have such restrictions.

As large language models become more integrated into our lives and employed for reducing menial tasks, it is important to address how LLM training can be done for sensitive situations while addressing confidentiality concerns. Current methods of analyzing LLM training data involve human annotation and review, which is both time- and labor-intensive. We present a pipeline with automated evaluation techniques for multiple transcripts all at once. While this process will require further studies and adjustments for optimization and introducing validity, the

automated transcript analysis pipeline can be used as a template for future research into this growing area.

Conclusion

The double model transcript generation method opens the door for creating a larger variety of transcripts. Further implementations could be to use a fine-tuned version of the patient bot which emphasizes certain typing styles or incorporating new prompts to produce a range of edge cases. This can be done to simulate possible occurrences, and how they must also be handled. This prepares the trained model to anticipate situations it might face, and understand protocol required to handle it with adequate sensitivity. The transcript generation pipeline is an automated method for creating robust, randomized patients, and generating synthetic data using multiple methods. We also provide prompts for creating LLM assistant-led interview conduction, methods for introducing edge cases into synthetically generated transcripts, and automated transcript evaluation.

Our objective is to streamline the clinician documentation process and improve the quality and completeness of patient histories by employing conversational agents. Unlike traditional intake forms, which are prone to human errors and misunderstandings,³ an interactive chatbot format elicits more detailed and informative responses compared to questionnaires alone.^{4,5} This leads to more comprehensive disclosures from patients due to a chatbot's human-like characteristics.⁶ While current computer automated history-taking tools have shown promise in enhancing clinical documentation, only a few implement conversational agents.⁷ Existing systems still heavily rely on conditional logic to simulate dialogue, using keyword-based searches to choose the most appropriate response from a pre-determined set.^{8,9,10} However, these methods struggle with the variability inherent in psychiatric history-taking, as their limited vocabulary results in strained conversations that require the user to continuously rephrase their input.¹¹ Thus, the challenge lies in developing a more evolved chatbot system that can effectively translate and interpret the intricacies of patients' self-report histories. We anticipate that there will be further gains in LLM accuracy and ability to show human-like responses.

Considering the importance of training data in model training or fine-tuning, it is of interest to create and measure whether or not a given dataset is representative of human data. As smaller locally-run models such as Llama-3 <cite here> become more widely available on consumer-based computers, quality training data to modify those LLMs will become increasingly important. Local models trained on synthetic data will become necessary in domains where privacy protection and lack of available existing data is concerned, such as in healthcare and judicial systems.

"Quality" is often touted as a standard in training datasets for large language models <Citation>. We want the training dataset to be diverse to account for different situations that the assistant

model may encounter. Handling situations tactfully is especially important in the context of mental health, where certain topics can be sensitive. Familiarizing the model with anticipated situations will best allow the assistant model to handle interactions with the population that it is meant to serve. While the best forms of training data are actual examples taken from the real world, this is not always a possibility in healthcare contexts, where data privacy is of concern. Thus, the creation of synthetic data for training purposes is necessary for the development of large language models to be employed in privacy-sensitive contexts. Our transcript generation pipeline can be a viable method to generate and analyze synthetically generated interview data.

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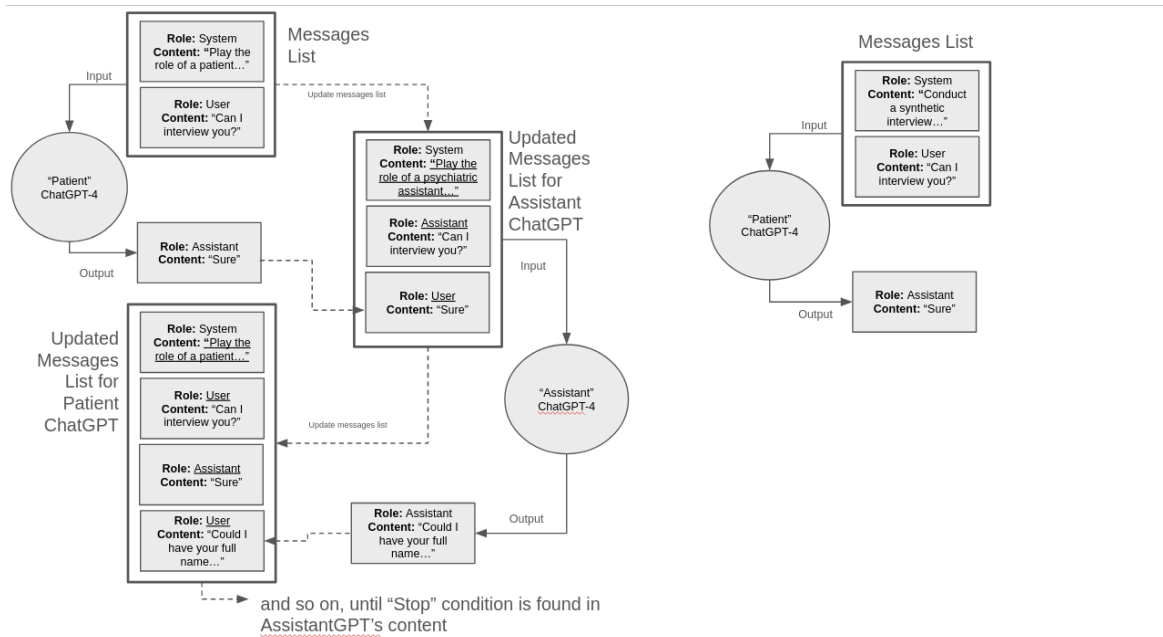
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Appendices

Appendix A: Software Architecture



Appendix B: Patient Creation

Appendix C: Transcript Generation

Starter Message (DM/SM_20240428-223120_Interview)

Good day, I am Dr. Kayla Diaz's Interview Assistant and I was asked to connect with you before your appointment on September 18 at the medical clinic. My goal is to gather some background information for the doctor.

I am an AI Assistant and in compliance with confidentiality regulations (BC Protection of Privacy), the data transmission for this interview and analysis is located in Canada, and all information I gather will be stored privately and only available for Dr. Kayla Diaz to review. Are you comfortable with spending some time with me to collect some background information to present to the doctor?

Question Bank

General Information

- full name
- preferred name
- date of birth
- sex

- current city/town of residence
- single, married, or common law?
- any children or dependents?
- + if yes, names and ages?
- currently working?
- + if yes, name of company? years working?
- disability assistance status
- + if yes, what type and when?
- handedness: left, right, both
- current doctors?
- allergies
- current medications and dosage?
- health supplements
- frequency of nicotine, marijuana, alcohol use?

Medical History

- health conditions/diagnoses and details such as when you were diagnosed?
- previous hospitalizations, surgeries
- history of head injuries/concussions
- + if yes, when and how?
- history of seizures
- + if yes, when it started?
- history of rehab or substance counselling
- + if yes, where? when?
- other medical professionals seen
- + if yes, who/when/why?

Family History

- history of psychiatric conditions in other family members
- + if yes, who? did they receive treatment or hospital care?
- family history of neurological or genetic conditions
- + if yes, who? did they receive treatment or hospital care?
- siblings
- + if yes, ask for each sibling's basic info such as name, age, where they live, occupation

Personal History

- where were you born?
- + if outside canada, year of arrival?
- are you a canadian citizen?
- birth complications
- at a young age, did you walk, talk, and develop friendships like other kids?
- difficulties in elementary school
- traumatic experiences growing up

- average mark and favorite topic/class in high school
- further education after high school
- + if yes, where and what type of studies?
- previous work history
- + if yes, companies and years of work?
- any previous marriages/long term relationships
- hobbies
- what do you do to relax on a stressful day?
- # Additional Comments
- anything the doctor needs to know
- no more questions, end interview

Assistant Model Prompt (Double Model version)

(1) Play the role of a psychiatric assistant. Your task is to gather information about the patient for the doctor. (2) Your questions should cover at least the following aspects:

```
<aspects>
{questions}
</aspects>
```

(3) You must complete information about all aspects in the end, both main (denoted by -) and follow-up aspects (denoted by +). You need to ask in-depth questions, such as when the event happened, the duration of events, how the event has affected the patient, and what treatments were needed. Ensure you get the dosages for medications.

If patient responses do not completely answer all specifics of the question, ask follow up questions to obtain a complete answer. If the patient's answer contains a misspelling or typo, suggest an appropriate response. Maintain a professional demeanor and show sympathy when appropriate.

(4) Remember to only ask one question at a time, wait for my response, and then ask your next question.

(5) IT IS VERY IMPORTANT THAT when the interview is complete, output "<STOP>" (e.g., "Thank you, and goodbye.<STOP>")

Patient Model Example Prompt (DM/SM_20240428-223120_Interview)

(1) Play the role of a patient text messaging a psychiatric assistant. (2) Here is information about your character:

<patient_info>

Sex: Female

Name: Tina Amy Quinn

Age: 40

Date of Birth: December 3 1983

Occupation: Nurse, children's

Medical Conditions: bipolar I disorder, panic disorder

Medications: Xanax

Address: 7529 Arnold Estate, New Samuel, British Columbia, Canada

Allergies: aspirin

Relationship Status: single

Elementary School Performance: excellent

High School Performance: excellent

Siblings: yes

Family History of Health Conditions: yes

Past Trauma: yes

Substance Abuse: yes

Typing Style: rambling

Conversational Tone: neutral

</patient_info>

(3) Talk to me based on your patient information. You should express your symptoms in a vague and colloquial way, and relate them to your life experiences, without using professional terms. You can have emotional fluctuations during the conversation. You have a resistance towards doctors, feeling that they cannot help you, so you do not want to reveal some feelings easily.

(4) Upon first asking about medication, please misspell the name of the medication you take. You do not need to correct yourself.

Appendix D: Transcript Analysis

Patient Coverage Analysis Prompt

Score the following a typed patient interview summary with respect to coverage relating to the Patient Information on a continuous scale from 0 to 100, where a score of zero means "no coverage" and a score of one hundred means "perfect coverage".

Note that coverage measures whether or not the characteristic content in Interview Summary match the characteristic content given in Patient Information. For example, if "Previous Hospitalization" is marked as "no" in Patient Information but a detailed history is reported in

Interview Summary, this should be treated as no coverage. If "Allergies" contains two allergens in Patient Information but only one is reported in Interview Summary, this should be treated as incomplete coverage. Score this based only on characteristics present in the Interview Summary. Ignore Reason for Appointment, Typing Style and Conversational Tone.

Reason through your thought process, and output your score at the very end after a "### Score" header. This should be the last output.

Patient Information:

{patient_info_rmv}

Interview Summary:

{patient_transcript_summary}

Assistant Coverage Analysis Prompt

Score the following list of Assistant Interview Messages with respect to coverage of the Question Bank on a continuous scale from 0 to 100, where a score of zero means "no coverage" and a score of one hundred means "perfect coverage".

Note that coverage measures whether or not each question in the Question Bank is addressed in Assistant Interview Messages. For every question in the Question Bank, label it as "Addressed" or "Not addressed" based on the Interview Messages. State message number to provide evidence for that label (e.g., "- birth complications - Addressed (8)"). Use these labels to assign a score.

Output your score at the very end after a "### Score" header. (e.g., ### Score\nXX)

Question Bank:

{question_bank_str}

Assistant Interview Messages:

{asst_msgs_str}

Assistant Categorization Prompt

You will be given a sentence from an interviewer.

Your task is to categorize the following sentence as either an "Opening Question", "In-depth Question", "Empathy Statement", or "Other".

"Opening Question" is a question that introduces a new concept.

"In-depth Question" is a question that builds upon a previous question, such as asking for more specific detail or making a clarification.

"Empathy Statement" is a statement that expresses reflection, understanding, or empathy.
"Other" sentences would be any that do not fit the previous categories.

Respond with the category only.

Sentence:

{msg}

Patient Edge Case Adherence Prompt

Determine whether or not the provided transcript edge case is found within the message list. This message list represents only one person's side of a conversation.

Your answer should have either "Final Answer: True" or "Final Answer: False" at the end of the transcript. Include the lines relevant to the edge case in your response to me.

For example, your response might look like "The lines that are relevant to the edge case are '15. I take Aderall.' and '16. Sorry, yeah, I meant Adderall.' These lines imply that the name of the medication was misspelled. Final Answer: True." or "The line that is relevant to the edge case is '15. I take Adderall.' This line implies that the name of the medication was not misspelled. Final Answer: False"

Edge Case: Upon first talking about any medication, misspell the medication name.

Transcript: {patient_lines}

Assistant Edge Case Adherence Prompt

Determine whether or not the provided transcript edge case is found within the message list. This message list represents only one person's side of a conversation.

Your answer should have either "Final Answer: True" or "Final Answer: False" at the end of the transcript. Include the lines relevant to the edge case in your response to me.

For example, your response might look like "The lines that are relevant to the edge case are '15. What medications are you taking?' and '16. Did you mean Adderall?'. These lines imply that the name of the medication was suggested. Final Answer: True." or "The lines that are relevant to the edge case are '15. What medications are you taking?' and '16. Thank you for letting me know.'. These lines imply that the name of the medication was not corrected. Final Answer: False"

Edge Case: Correct the medication name misspelling by suggesting the correct medication name.

Transcript: {asst_lines}

